

```
In [1]: !pipx inject notebook numpy pandas matplotlib seaborn
```

```
injected package numpy into venv notebook
done! ✨ 🌟 ✨
injected package pandas into venv notebook
done! ✨ 🌟 ✨
injected package matplotlib into venv notebook
done! ✨ 🌟 ✨
injected package seaborn into venv notebook
done! ✨ 🌟 ✨
```

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

<h1 id="Iris-DataSet">**Iris DataSet**</h1><p>Iris dataset has the follwing features</p>

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

<p>we have to use them to classify them to classes such as:</p>

- Iris-setosa
- Iris-versicolor
- Iris-virginica

<h1 id="1.-Loading-the-dataset">1. Loading the dataset</h1>

```
ascii
Parameters
-----
filepath : str
    Path to dataset.

column_names : list, optional
    List of column names.

separator : str
    Delimiter used in the file.

Returns
-----
DataFrame
```

```
In [3]: def load_dataset(filepath,column_names=None,separator=","):
df=pd.read_csv(filepath,names=column_names,sep=separator)
df.columns = df.columns.str.strip()
```

```
return df
```

<h1 id="2.-Dataset-Information">2. Dataset Information</h1><p>Displays basic information about the data</p>

- shape
- columns
- info
- head

```
In [4]: def dataset_info(df):
        print("-" * 50)
        print("DATASET INFORMATION")
        print("-"*50)

        print("\n Shape: ",df.shape)
        print("\nColumns: ")
        print(df.columns.tolist())

        print("\nData Types: ")
        print(df.dtypes)

        print("\nMemory information")
        df.info()

        print("First 5 elements")
        print(df.head())
```

<h1 id="3.Missing-Values-and-duplicate-values">3.Missing Values and duplicate values</h1>

```
In [5]: def missing_values(df):
        missing=df.isnull().sum()
        percent=(missing/len(df) )*100

        result=pd.DataFrame(
        {"missing":missing,
        "percent":percent
        })

        print(result)

        if (missing.sum() ==0):
            print("There is no missing value!")
        else:
            print("Missing Values detected")
        return result

def duplicate_info(df):
    duplicates = df.duplicated().sum()
    print("\nThe duplicate rows are : ",duplicates);

    if(duplicates ==0):
        print("No duplicates detected!")
```

```
else:
    print("Duplicates are detected")

return duplicates
```

```
In [6]: column_names = [
        "sepal_length",
        "sepal_width",
        "petal_length",
        "petal_width",
        "class"
    ]

df = load_dataset(
    "iris/iris.data",
    column_names
)

dataset_info(df)

missing_values(df)

_ = duplicate_info(df)
```

 DATASET INFORMATION

Shape: (150, 5)

Columns:

['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'class']

Data Types:

sepal_length float64

sepal_width float64

petal_length float64

petal_width float64

class str

dtype: object

Memory information

<class 'pandas.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal_length	150 non-null	float64
1	sepal_width	150 non-null	float64
2	petal_length	150 non-null	float64
3	petal_width	150 non-null	float64
4	class	150 non-null	str

dtypes: float64(4), str(1)

memory usage: 6.0 KB

First 5 elements

	sepal_length	sepal_width	petal_length	petal_width	class
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

	missing	percent
sepal_length	0	0.0
sepal_width	0	0.0
petal_length	0	0.0
petal_width	0	0.0
class	0	0.0

There is no missing value!

The duplicate rows are : 3

Duplicates are detected

<h2 id="Observation:">Observation:</h2>

- The dataset contains 150 entries with the 5 column
- The first four are the numerical column and the other one is the class to be predicted
- No missing data is found
- Three duplicate values is detected!

<h1 id="4.-EDA-for-a-dataset">4. EDA for a dataset</h1>

```
In [7]: def summary_statistics(df):
        print("\nSummary statsitics")
        print(df.describe())

        def class_distribution(df,target_column):
            print("\nClass Distribution")

            counts=df[target_column].value_counts()
            counts.plot(kind="bar")
            plt.xlabel(target_column)
            plt.ylabel("Count")
            plt.title(f"Class Distribution of {target_column}")
            plt.show()

            return counts
```

```
In [8]: summary_statistics(df)
```

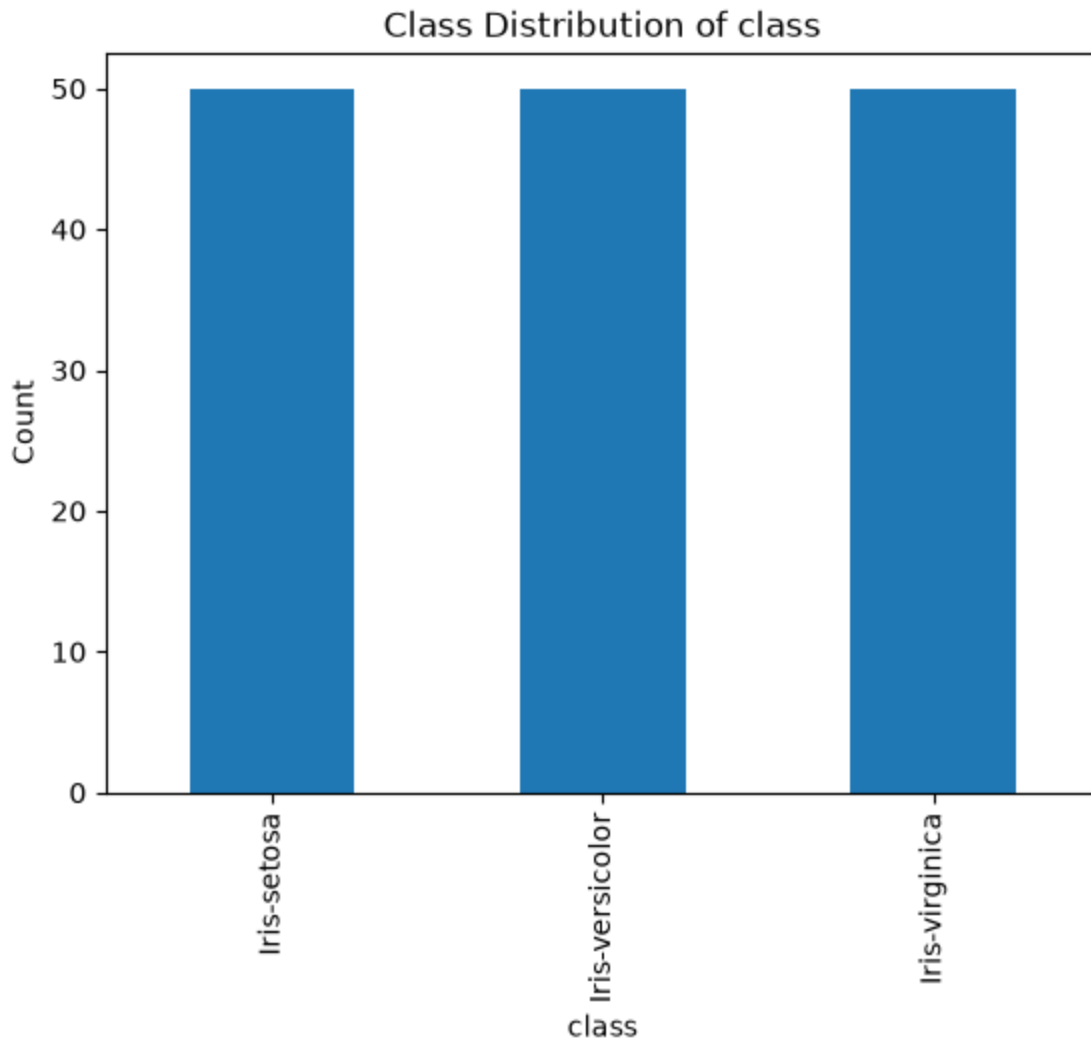
```
Summary statsitics
      sepal_length  sepal_width  petal_length  petal_width
count      150.000000      150.000000      150.000000      150.000000
mean         5.843333         3.054000         3.758667         1.198667
std          0.828066         0.433594         1.764420         0.763161
min          4.300000         2.000000         1.000000         0.100000
25%          5.100000         2.800000         1.600000         0.300000
50%          5.800000         3.000000         4.350000         1.300000
75%          6.400000         3.300000         5.100000         1.800000
max          7.900000         4.400000         6.900000         2.500000
```

<h2 id="Observation-(summary):">Observation (summary):</h2>

- petal length has highest deviation from its standaed deviation
- Overview of each numerical stat is given

```
In [9]: class_distribution(df,"class")
```

Class Distribution



```
Out[9]: class
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64
```

<h2 id="Observation---Class-Distribution">Observation - Class Distribution</h2>

- Each Iris species has 50 samples
- The dataset is balanced across in all the three classes
- A balanced dataset helps prevent bias toward any particular class during model training.

```
In [10]: def plot_histogram(df,bins=10):
          numeric_df=df.select_dtypes("number")

          numeric_df.hist(bins=bins)
          plt.suptitle("Histogram of numerical features")
          plt.tight_layout()
          plt.show()

          def plot_boxplot(df):
              numeric_df = df.select_dtypes("number")
```

```

for col in numeric_df.columns:
    plt.boxplot(numeric_df[col])
    plt.title("BoxPlot of "+str(col))
    plt.show()
def plot_heatmap(df):
    numeric_df = df.select_dtypes("number")
    corr = numeric_df.corr()
    sns.heatmap(corr,annot=True)
    plt.title("Heatmap !")
    plt.show()

def plot_scatterplot(df,target_column=None):

    if (target_column is not None):
        sns.pairplot(df,hue=target_column)
    else:
        sns.pairplot(df)
    plt.suptitle("Pairplot")
    plt.tight_layout()
    plt.show()

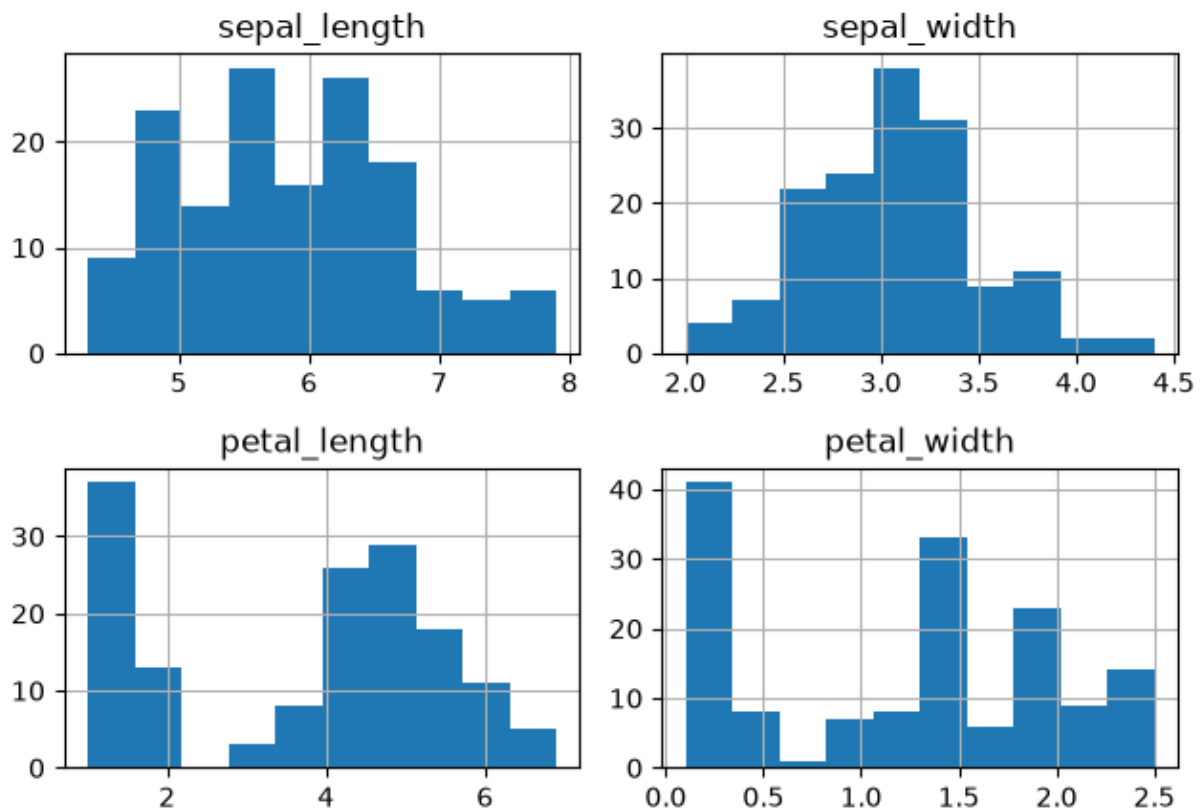
```

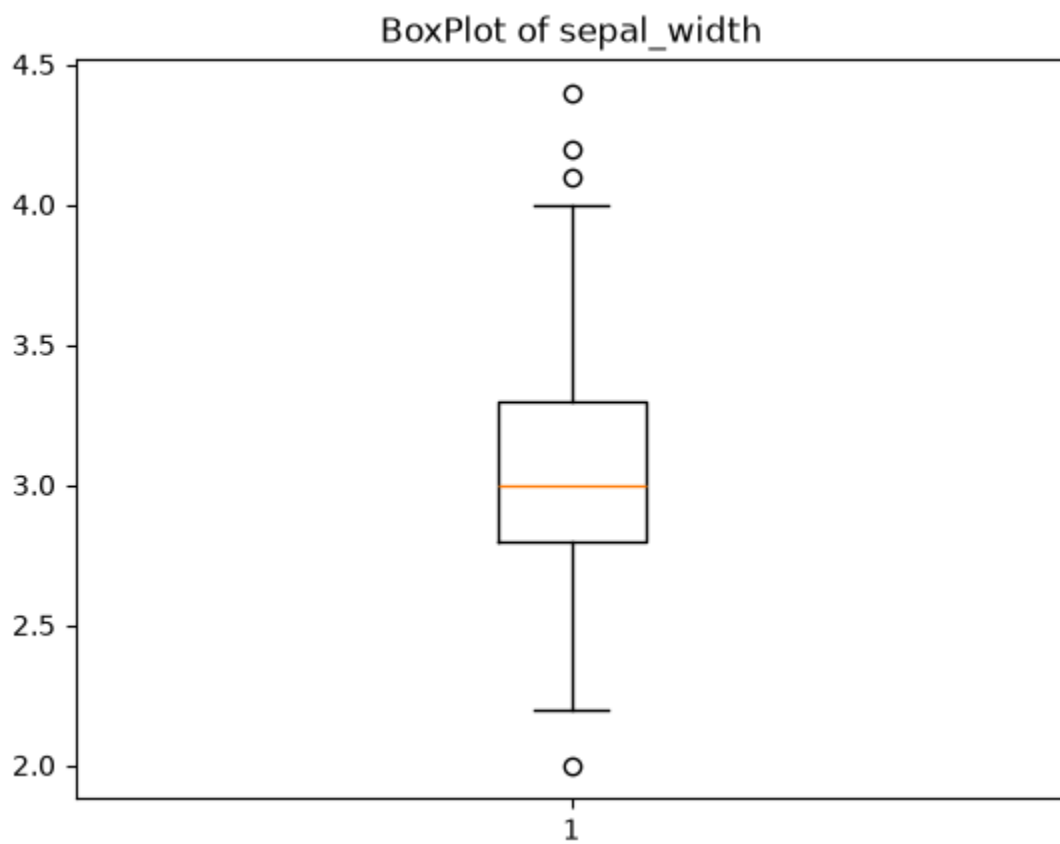
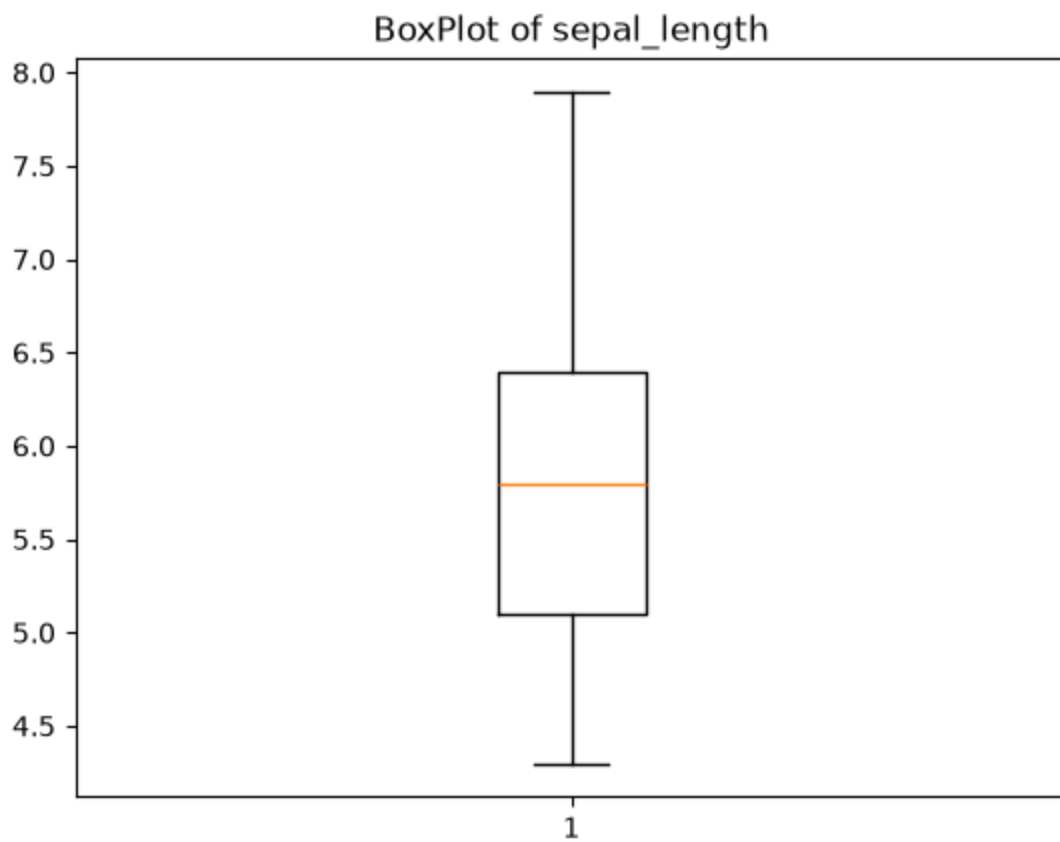
```

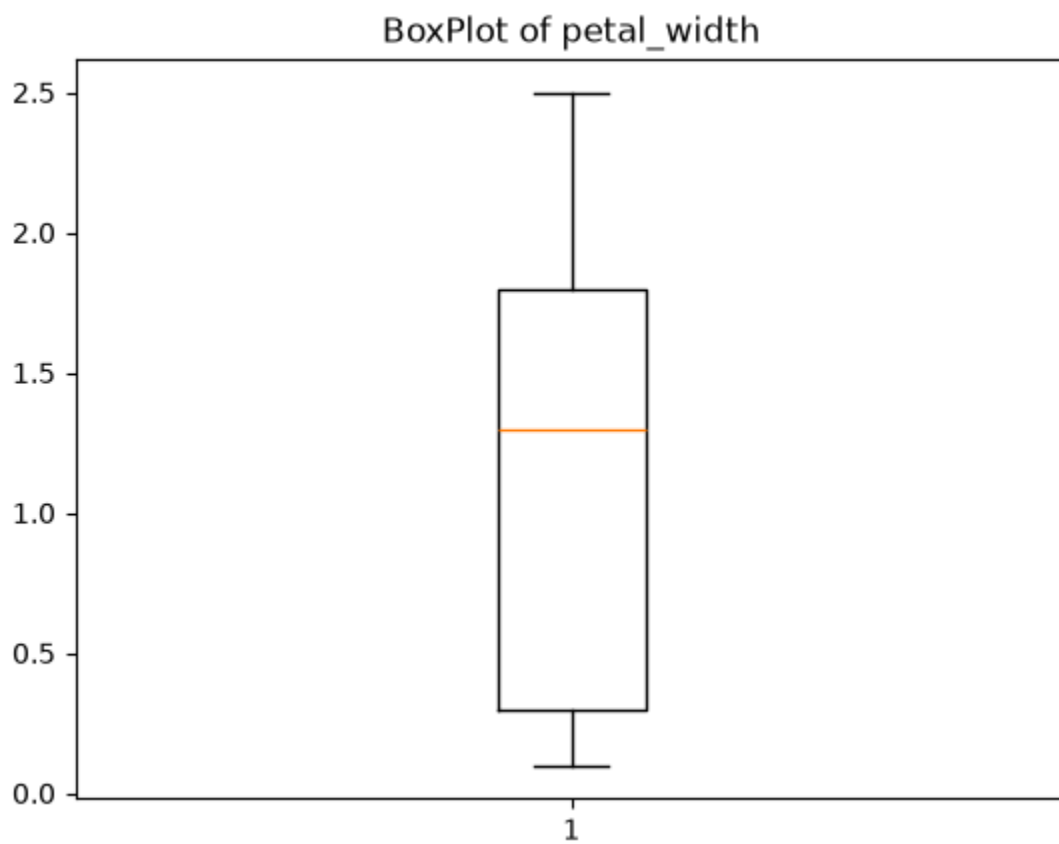
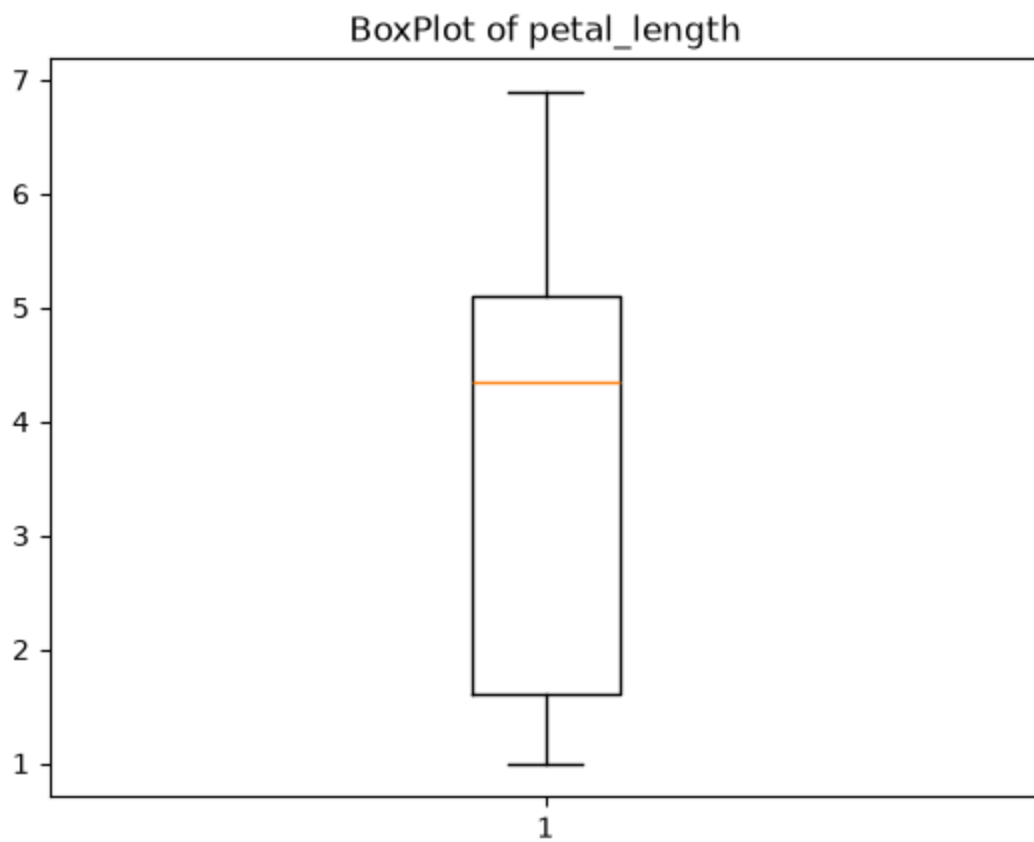
In [11]: plot_histogram(df)
         plot_boxplot(df)
         plot_heatmap(df)
         plot_scatterplot(df)

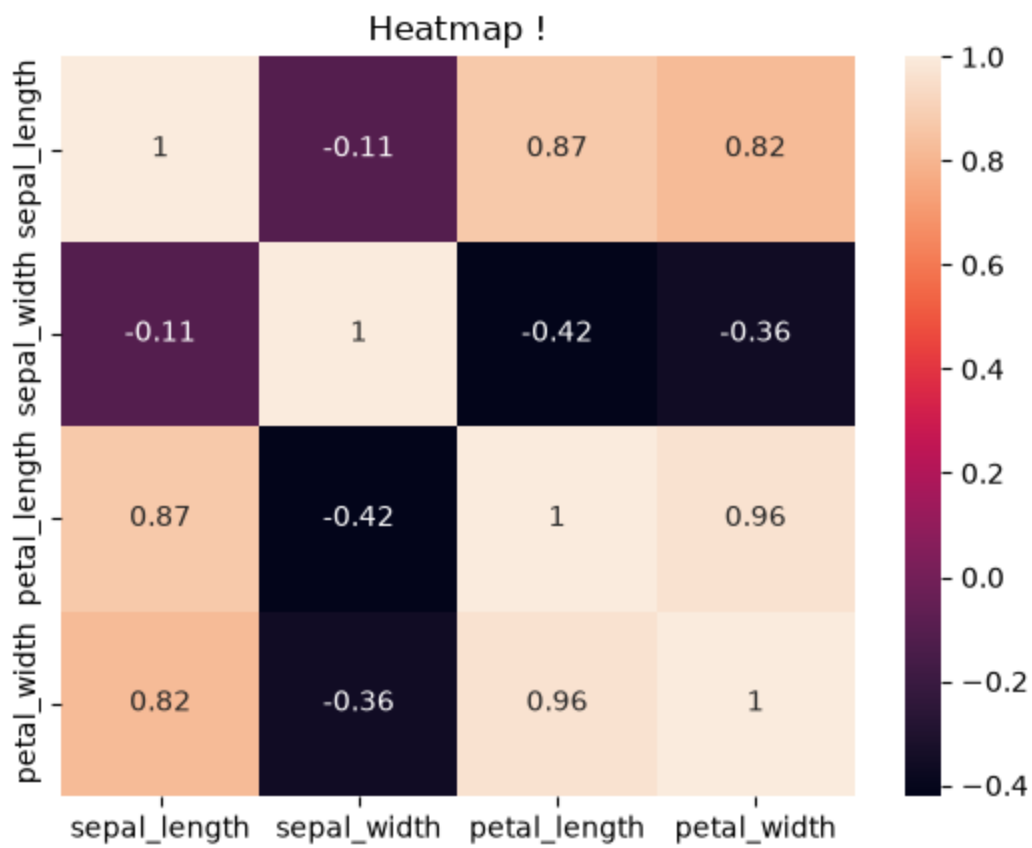
```

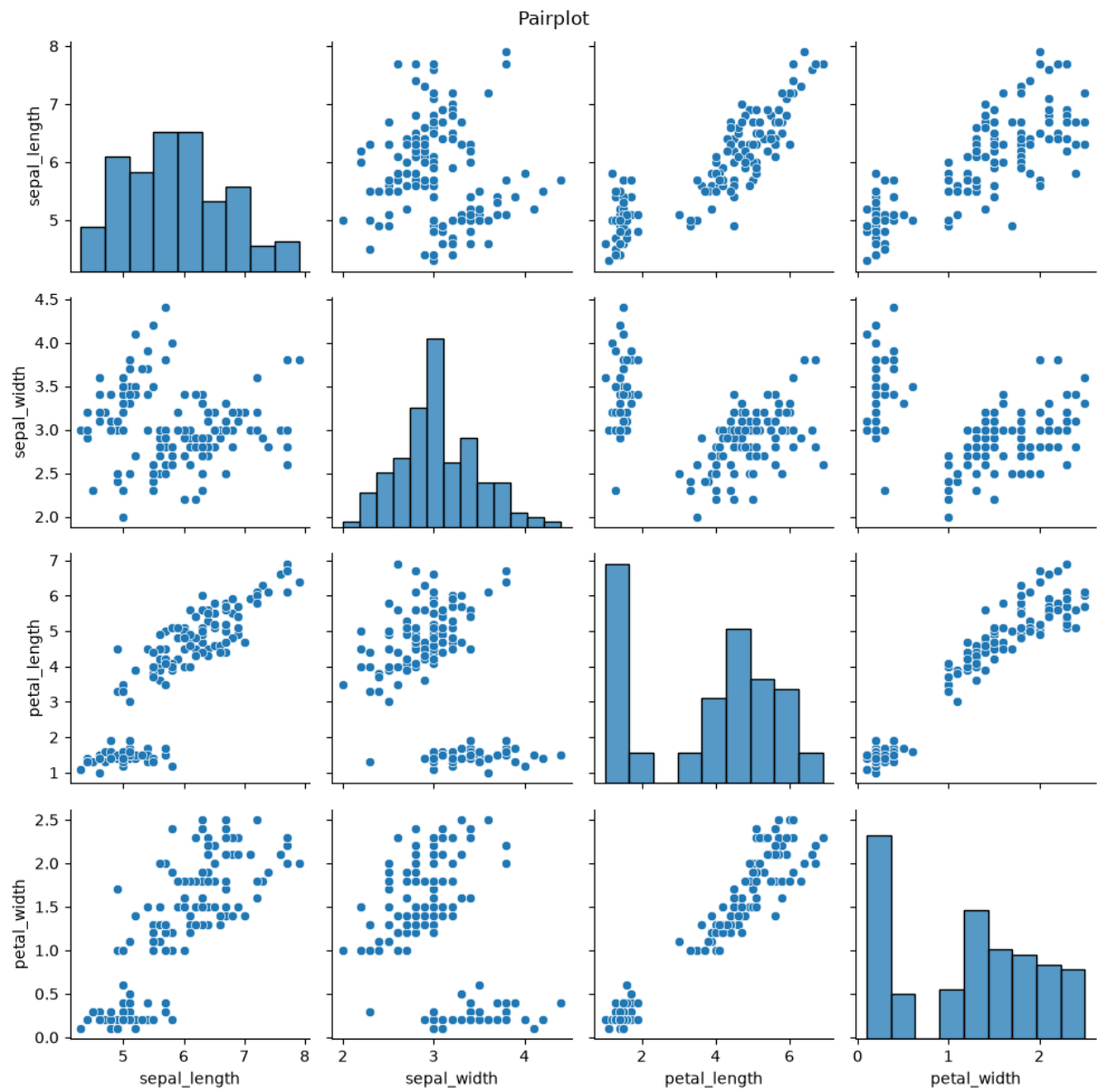
Histogram of numerical features











<h2 id="Observation---Histogram">Observation - Histogram</h2>

- Sepal_length and sepal_width are roughly unimodal, centered around their respective means.
- Petal_length and petal_width are clearly **bimodal**, with a distinct low cluster (petal_length ~1–2, petal_width ~0–0.3) separated from a second, wider cluster (petal_length ~3–7, petal_width ~1–2.5).
- This bimodal split strongly suggests one species (setosa) is separable from the other two using petal measurements alone.

<h2 id="Observation---Box-Plot">Observation - Box Plot</h2>

- Sepal_length, petal_length, and petal_width show **no outliers**.
- Sepal_width is the only feature with outliers, with points above **4.0** and below **2.0**.
- Petal_length has the **widest interquartile range (1.6–5.1)**, consistent with it spanning two distinct clusters rather than one.

<h2 id="Observation---Correlation-Heatmap">Observation - Correlation Heatmap</h2>

- Petal length and petal width show a very strong positive correlation (approximately 0.96).
- Sepal length is also strongly positively correlated with petal length and petal width.
- Sepal width shows weak to moderate negative correlations with the petal measurements.
- These relationships indicate that petal measurements are the most informative features for distinguishing Iris species.

<h2 id="Observation---Pair-Plot">Observation - Pair Plot</h2>

- The pair plot displays pairwise relationships between all numerical features in the dataset.
- The diagonal plots show the distribution of each individual feature using histograms.
- Several scatter plots exhibit natural clustering of data points, indicating the presence of distinct groups within the dataset.
- Petal length and petal width show the clearest clustering pattern and a strong positive linear relationship.
- Sepal length also exhibits a positive relationship with both petal length and petal width.
- Sepal width shows comparatively weaker relationships and greater overlap among the observations.
- The pair plot suggests that petal-related features are more effective in distinguishing different flower samples than sepal-related features.

<h1 id="2.Loan-approval-Dataset">2.Loan approval Dataset</h1><p>Features: Basic

deails of the loaner(like education,income,loan term,cibil score,etc)</p>

<p>Prediction:</p>

- Loan amount

```
In [12]: '''  
Loading the dataset  
'''  
df = load_dataset("LoanAmount/loan_approval_dataset.csv")
```

```
In [13]: '''  
basic data analysis  
'''  
dataset_info(df)  
  
missing_values(df)  
  
duplicate_info(df)  
  
summary_statistics(df)
```

 DATASET INFORMATION

Shape: (4269, 13)

Columns:

['loan_id', 'no_of_dependents', 'education', 'self_employed', 'income_annum',
 'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value', 'co
 mmercial_assets_value', 'luxury_assets_value', 'bank_asset_value', 'loan_sta
 tus']

Data Types:

```
loan_id                int64
no_of_dependents      int64
education             str
self_employed         str
income_annum          int64
loan_amount           int64
loan_term             int64
cibil_score           int64
residential_assets_value int64
commercial_assets_value int64
luxury_assets_value   int64
bank_asset_value       int64
loan_status           str
dtype: object
```

Memory information

<class 'pandas.DataFrame'>

RangeIndex: 4269 entries, 0 to 4268

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	loan_id	4269 non-null	int64
1	no_of_dependents	4269 non-null	int64
2	education	4269 non-null	str
3	self_employed	4269 non-null	str
4	income_annum	4269 non-null	int64
5	loan_amount	4269 non-null	int64
6	loan_term	4269 non-null	int64
7	cibil_score	4269 non-null	int64
8	residential_assets_value	4269 non-null	int64
9	commercial_assets_value	4269 non-null	int64
10	luxury_assets_value	4269 non-null	int64
11	bank_asset_value	4269 non-null	int64
12	loan_status	4269 non-null	str

dtypes: int64(10), str(3)

memory usage: 433.7 KB

First 5 elements

	loan_id	no_of_dependents	education	self_employed	income_annum	\
0	1	2	Graduate	No	9600000	
1	2	0	Not Graduate	Yes	4100000	
2	3	3	Graduate	No	9100000	
3	4	3	Graduate	No	8200000	
4	5	5	Not Graduate	Yes	9800000	

	loan_amount	loan_term	cibil_score	residential_assets_value	\
0	29900000	12	778	2400000	
1	12200000	8	417	2700000	
2	29700000	20	506	7100000	
3	30700000	8	467	18200000	
4	24200000	20	382	12400000	

	commercial_assets_value	luxury_assets_value	bank_asset_value	loan_status
0	17600000	22700000	8000000	Approved
1	2200000	8800000	3300000	Rejected
2	4500000	33300000	12800000	Rejected
3	3300000	23300000	7900000	Rejected
4	8200000	29400000	5000000	Rejected

	missing	percent
loan_id	0	0.0
no_of_dependents	0	0.0
education	0	0.0
self_employed	0	0.0
income_annum	0	0.0
loan_amount	0	0.0
loan_term	0	0.0
cibil_score	0	0.0
residential_assets_value	0	0.0
commercial_assets_value	0	0.0
luxury_assets_value	0	0.0
bank_asset_value	0	0.0
loan_status	0	0.0

There is no missing value!

The duplicate rows are : 0

No duplicates detected!

Summary statistics

	loan_id	no_of_dependents	income_annum	loan_amount	loan_term
count	4269.000000	4269.000000	4.269000e+03	4.269000e+03	4269.000000
mean	2135.000000	2.498712	5.059124e+06	1.513345e+07	10.900445
std	1232.498479	1.695910	2.806840e+06	9.043363e+06	5.709187
min	1.000000	0.000000	2.000000e+05	3.000000e+05	2.000000
25%	1068.000000	1.000000	2.700000e+06	7.700000e+06	6.000000
50%	2135.000000	3.000000	5.100000e+06	1.450000e+07	10.000000
75%	3202.000000	4.000000	7.500000e+06	2.150000e+07	16.000000

max	4269.000000	5.000000	9.900000e+06	3.950000e+07	20.000000
0					

	cibil_score	residential_assets_value	commercial_assets_value	\
count	4269.000000	4.269000e+03	4.269000e+03	
mean	599.936051	7.472617e+06	4.973155e+06	
std	172.430401	6.503637e+06	4.388966e+06	
min	300.000000	-1.000000e+05	0.000000e+00	
25%	453.000000	2.200000e+06	1.300000e+06	
50%	600.000000	5.600000e+06	3.700000e+06	
75%	748.000000	1.130000e+07	7.600000e+06	
max	900.000000	2.910000e+07	1.940000e+07	

	luxury_assets_value	bank_asset_value
count	4.269000e+03	4.269000e+03
mean	1.512631e+07	4.976692e+06
std	9.103754e+06	3.250185e+06
min	3.000000e+05	0.000000e+00
25%	7.500000e+06	2.300000e+06
50%	1.460000e+07	4.600000e+06
75%	2.170000e+07	7.100000e+06
max	3.920000e+07	1.470000e+07

Observation - Dataset Information

- The dataset contains **4,269 records** and **13 features**.
- There are **10 numerical features** and **3 categorical features** (education, self_employed, and loan_status).
- All columns have valid data types and are suitable for further analysis.
- The dataset contains applicant information, financial details, and loan approval status.

Observation - Missing Values

- No missing values are present in any of the dataset columns.
- Therefore, no data cleaning or imputation is required before analysis.

Observation - Duplicate Values

- No duplicate records were found in the dataset.
- Every record represents a unique loan application.

Observation - Summary Statistics

- Applicant income varies in a wide range, this shows clients are from different economic background
- Loan amounts also cover a broad range, suggesting loans of varying sizes.
- CIBIL scores range from **300 to 900**, representing applicants with different creditworthiness.
- Loan terms vary between **2 and 20 years**.
- An unusual negative value is observed in the residential asset column, which may

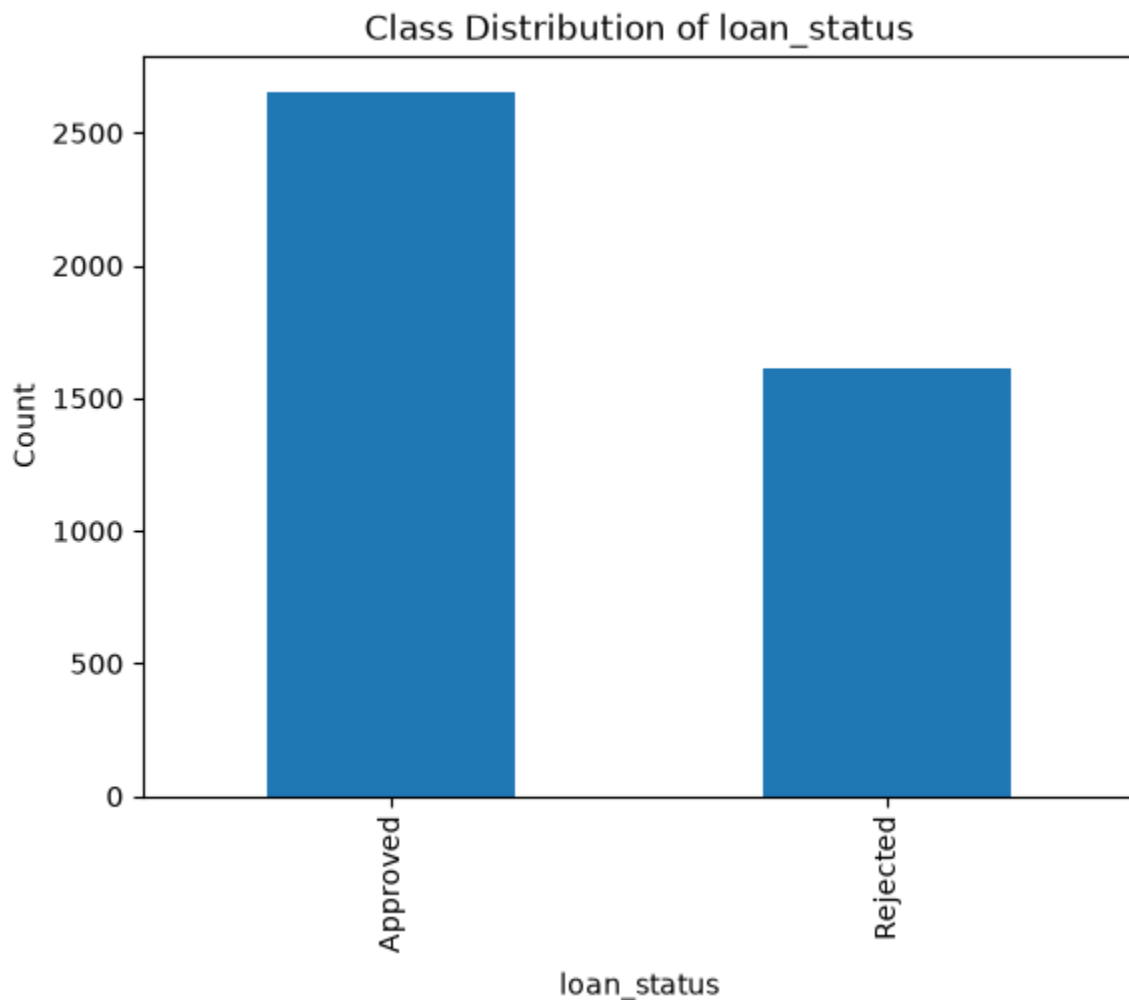
indicate a data entry error or special case that should be examined during preprocessing.

<h2 id="EDA">EDA</h2>

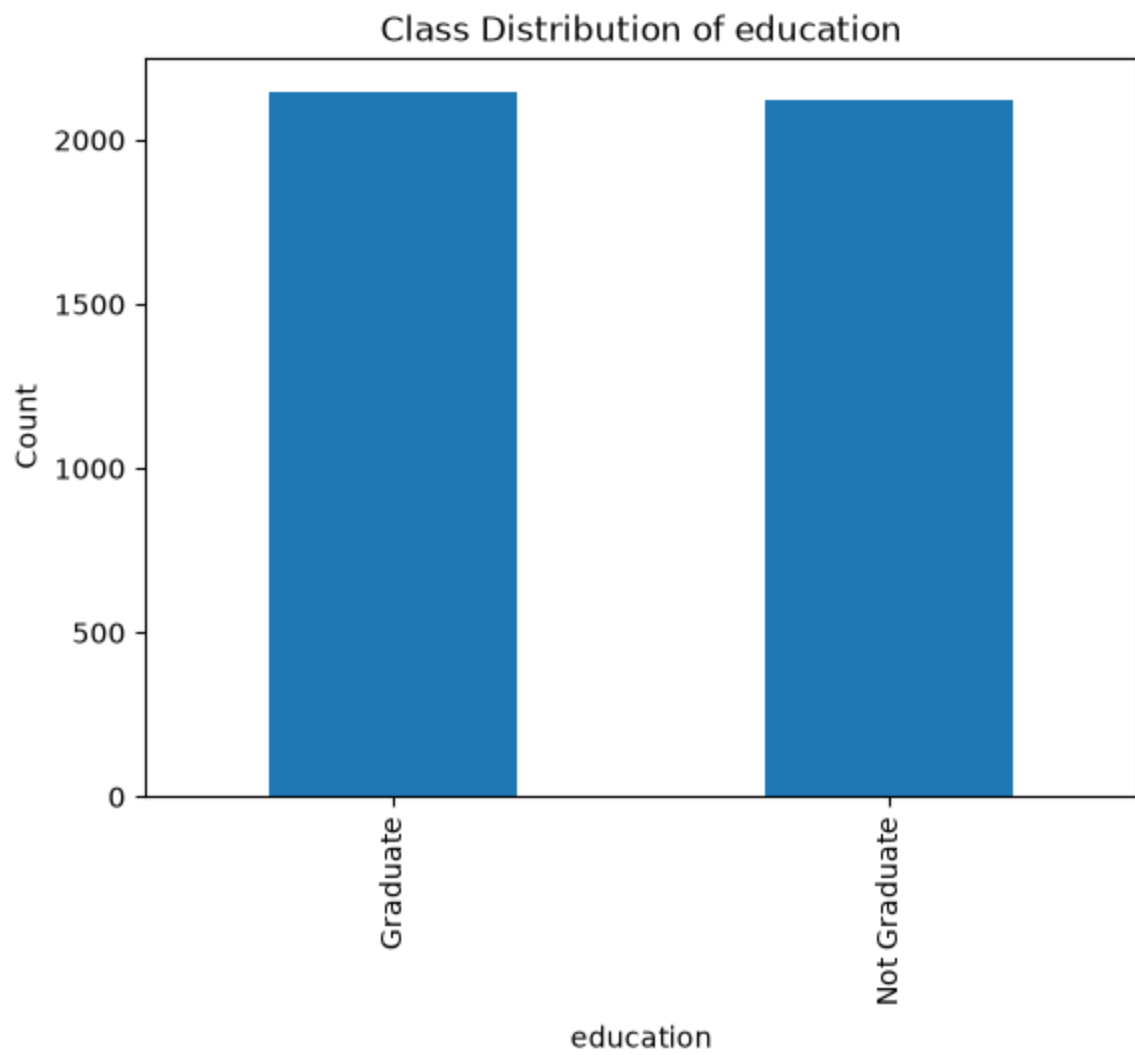
```
In [14]: class_distribution(df, "loan_status")
class_distribution(df, "education")
class_distribution(df, "self_employed")
plot_histogram(df)
plot_boxplot(df)
plot_heatmap(df)
important_features = [
    "income_annum",
    "loan_amount",
    "cibil_score",
    "loan_term",
    "loan_status"
]

plot_scatterplot(df[important_features], "loan_status")
```

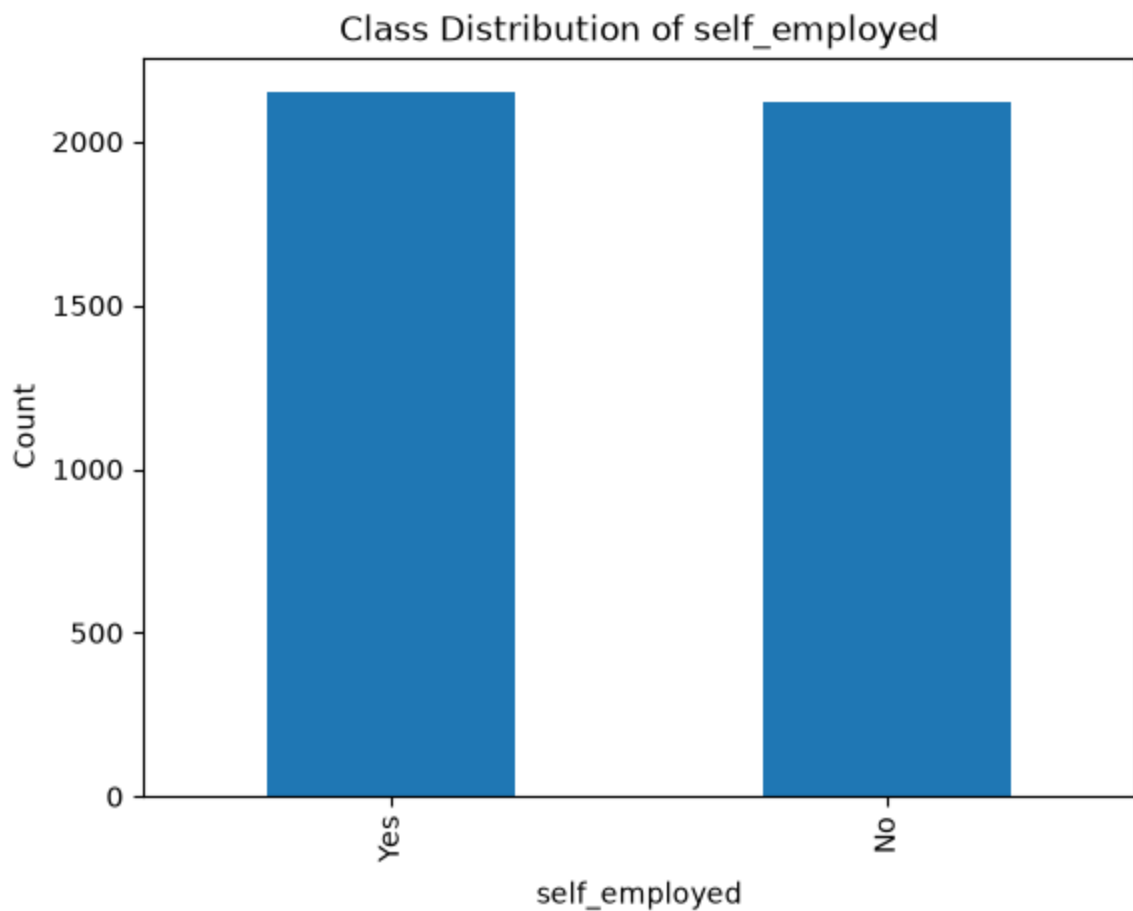
Class Distribution



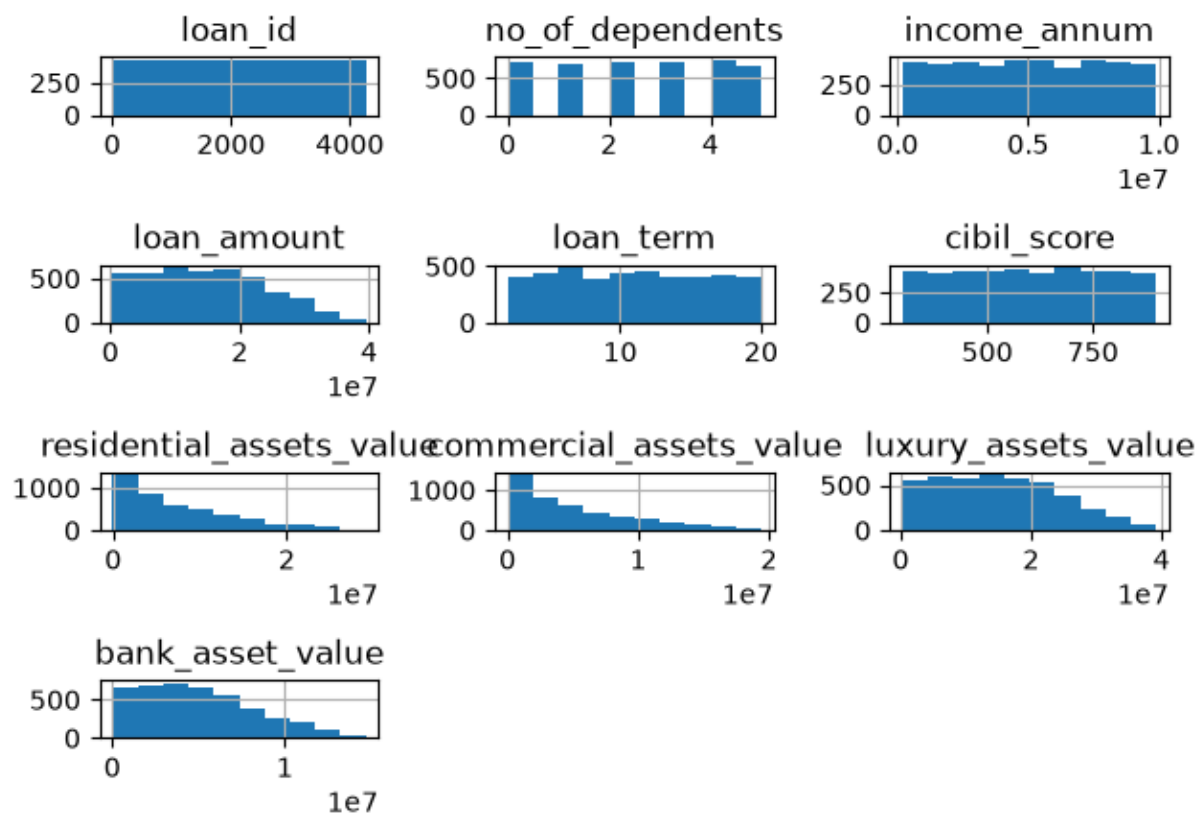
Class Distribution

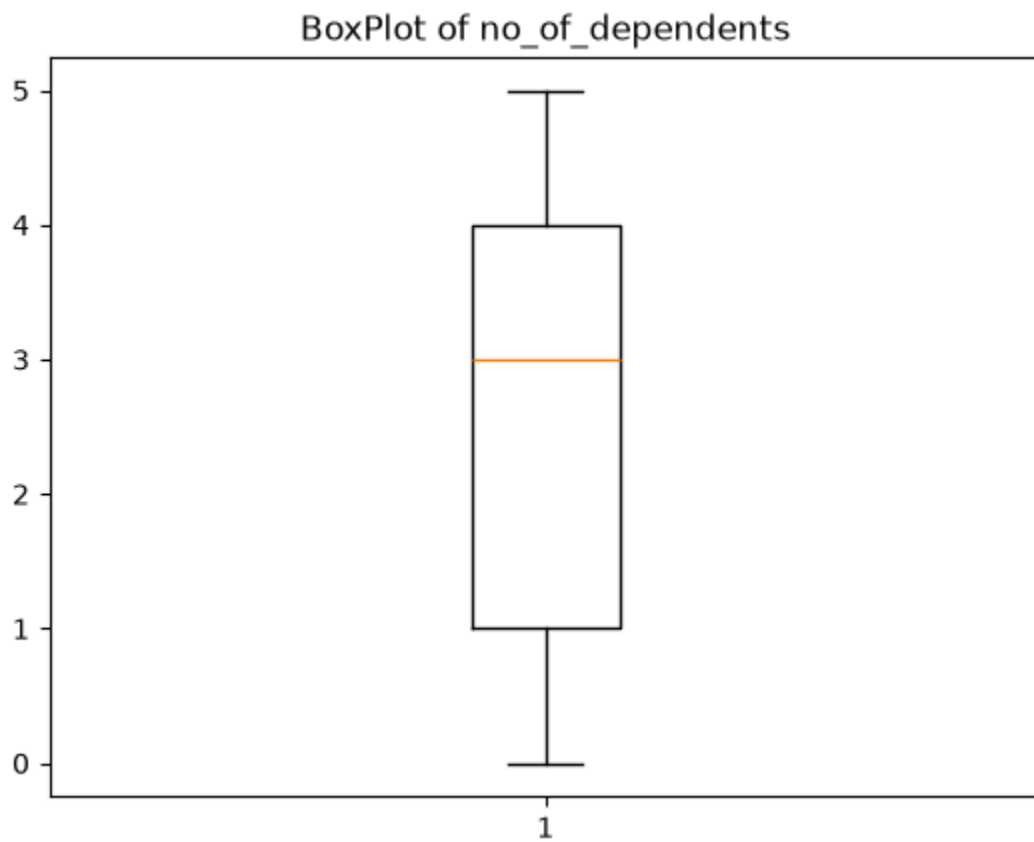
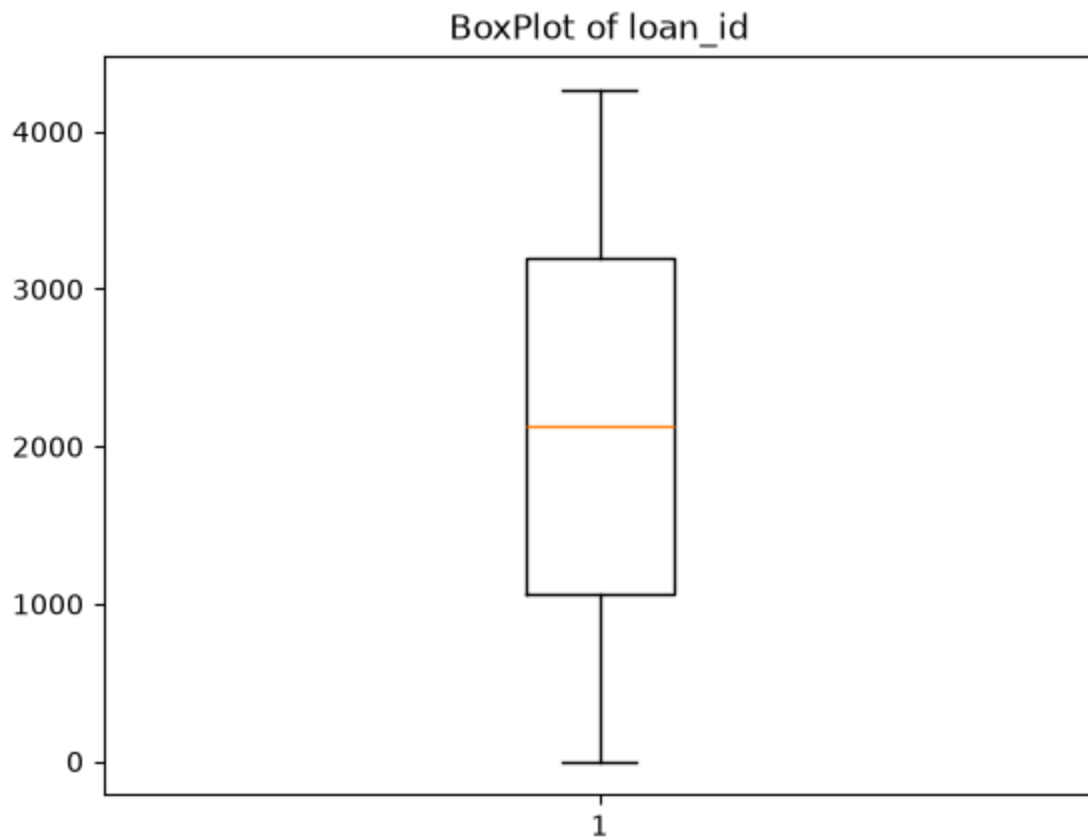


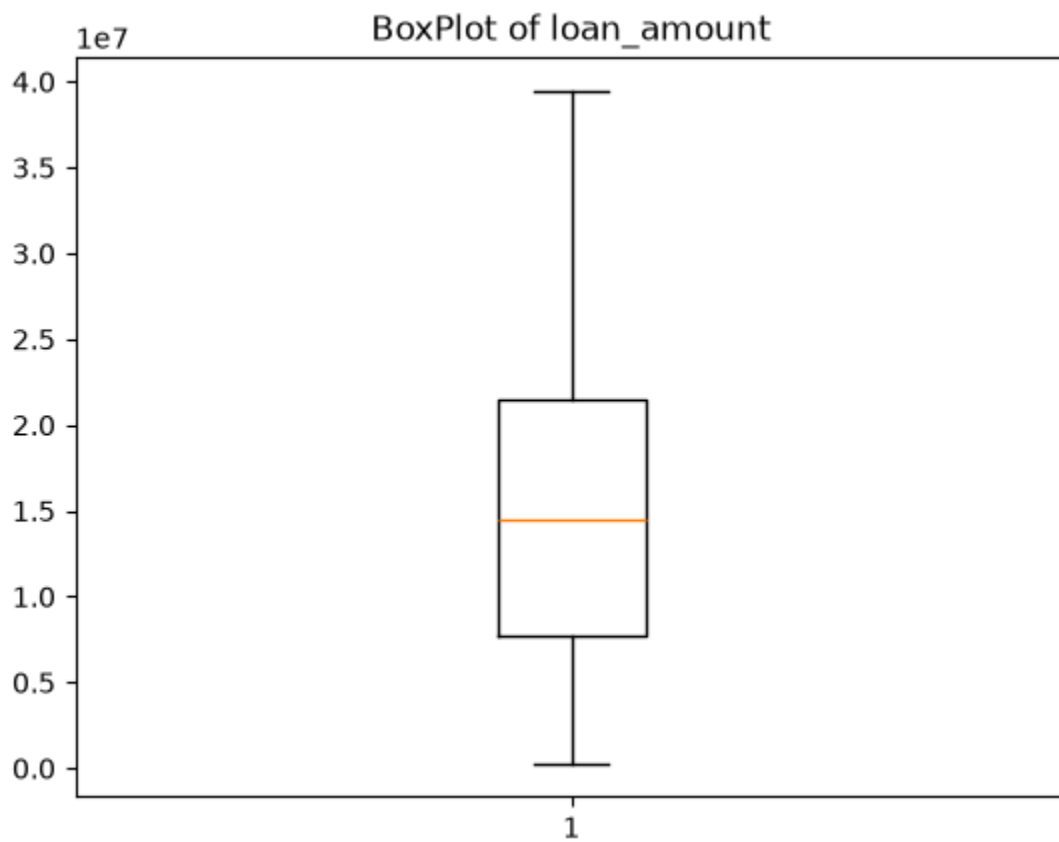
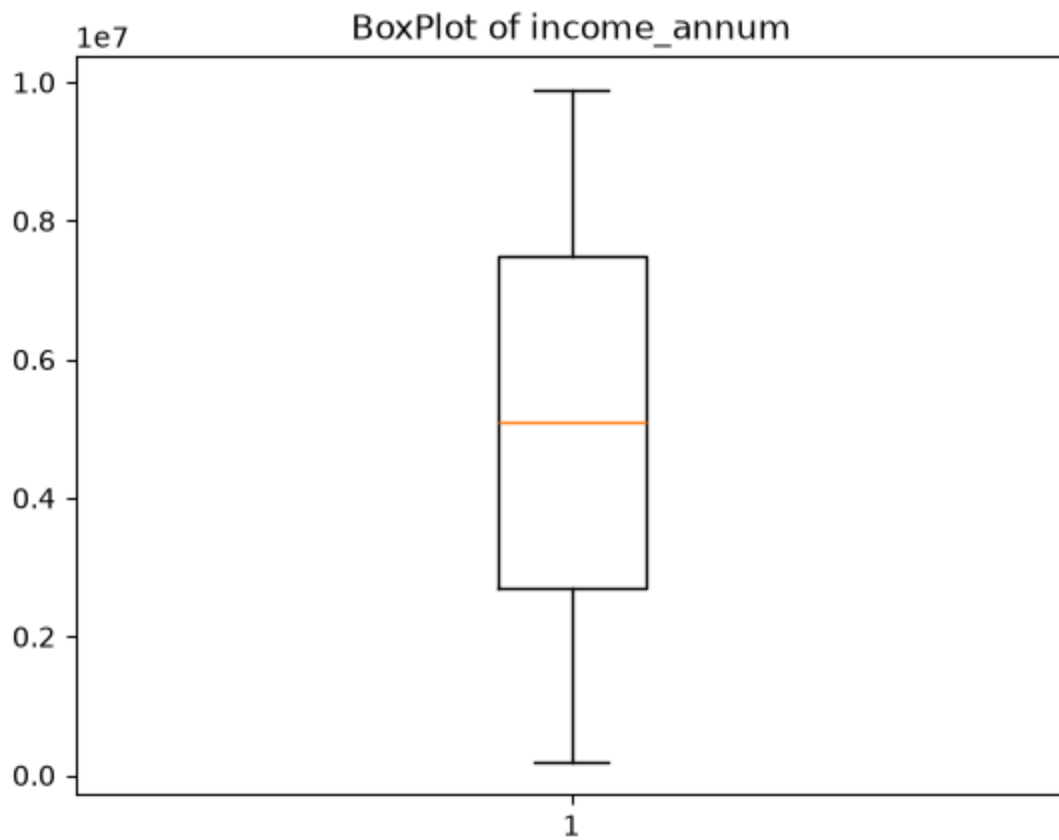
Class Distribution

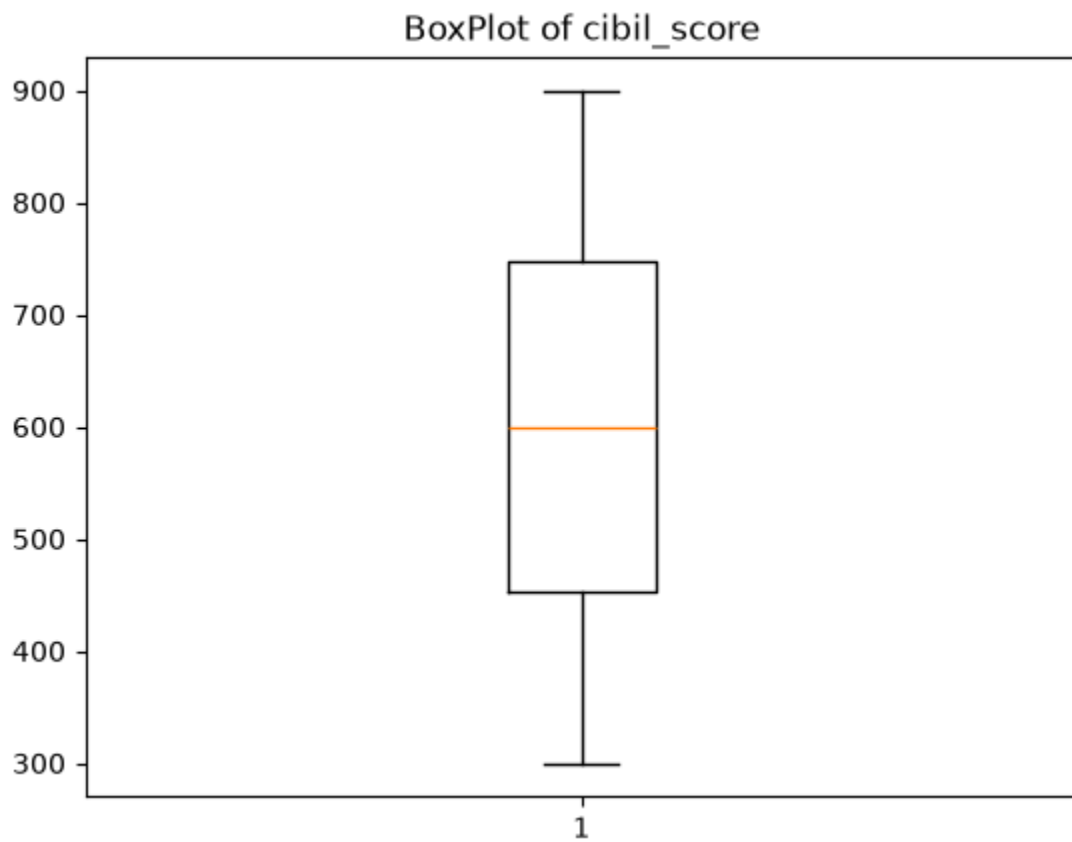
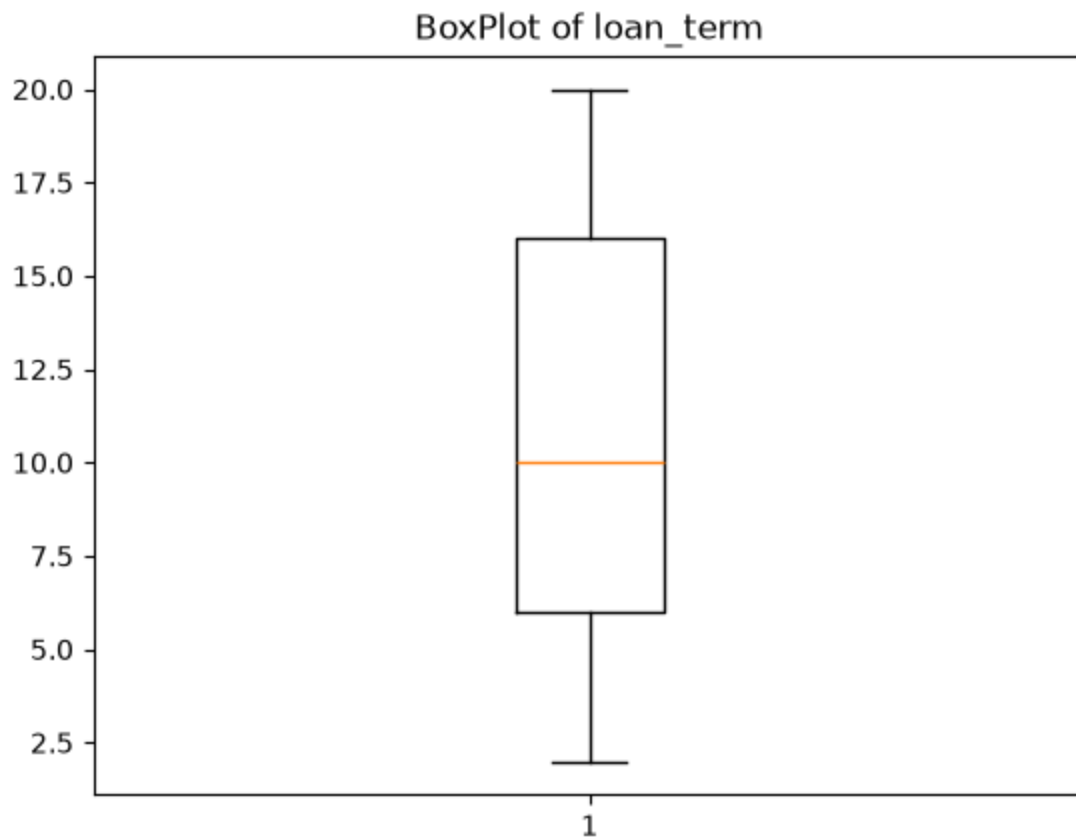


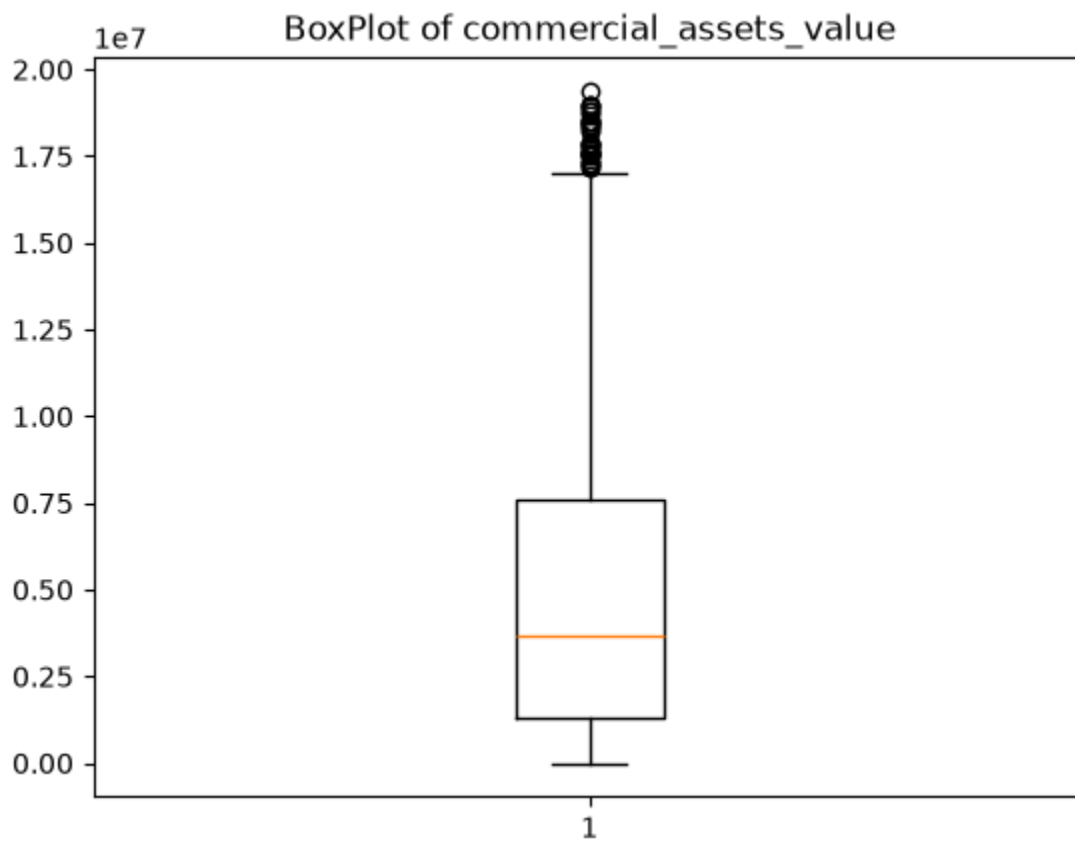
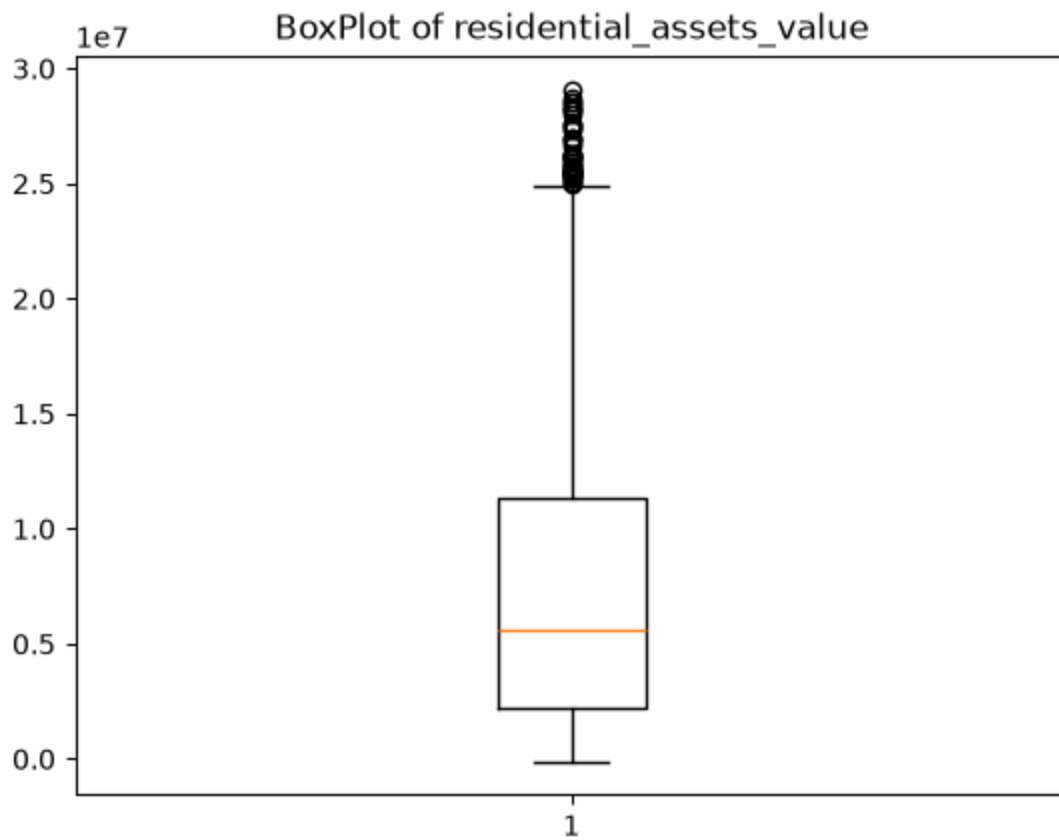
Histogram of numerical features

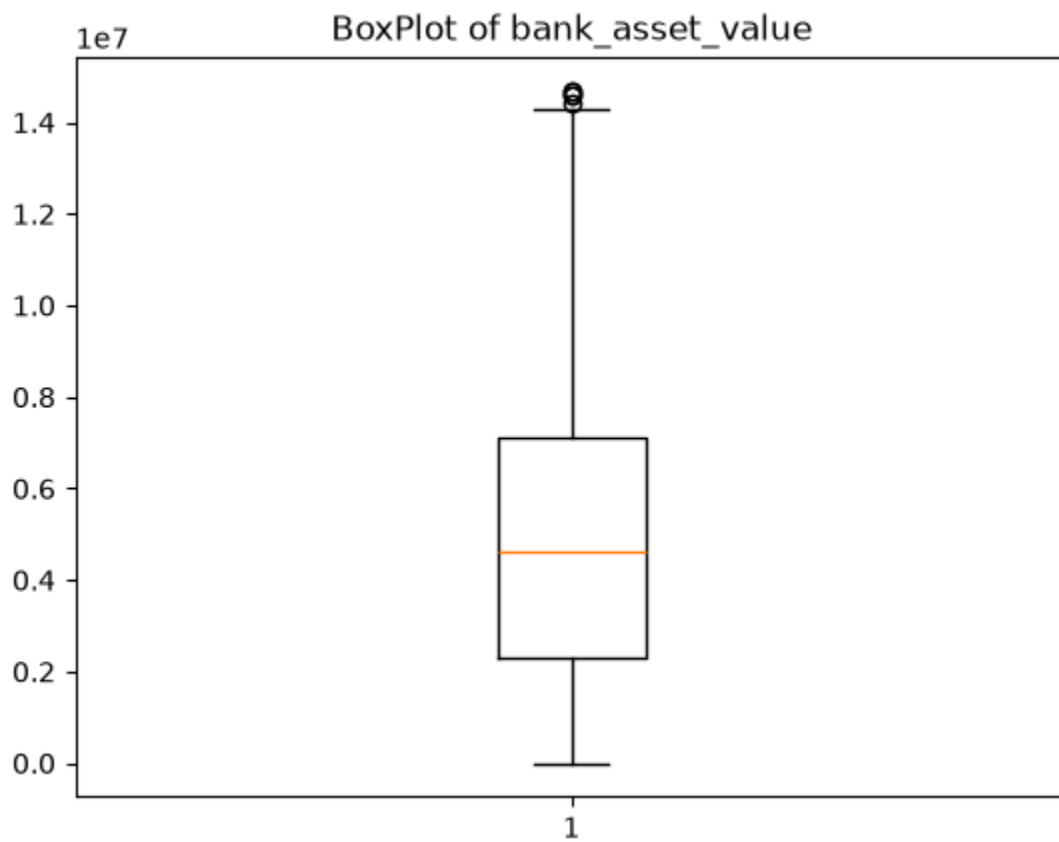
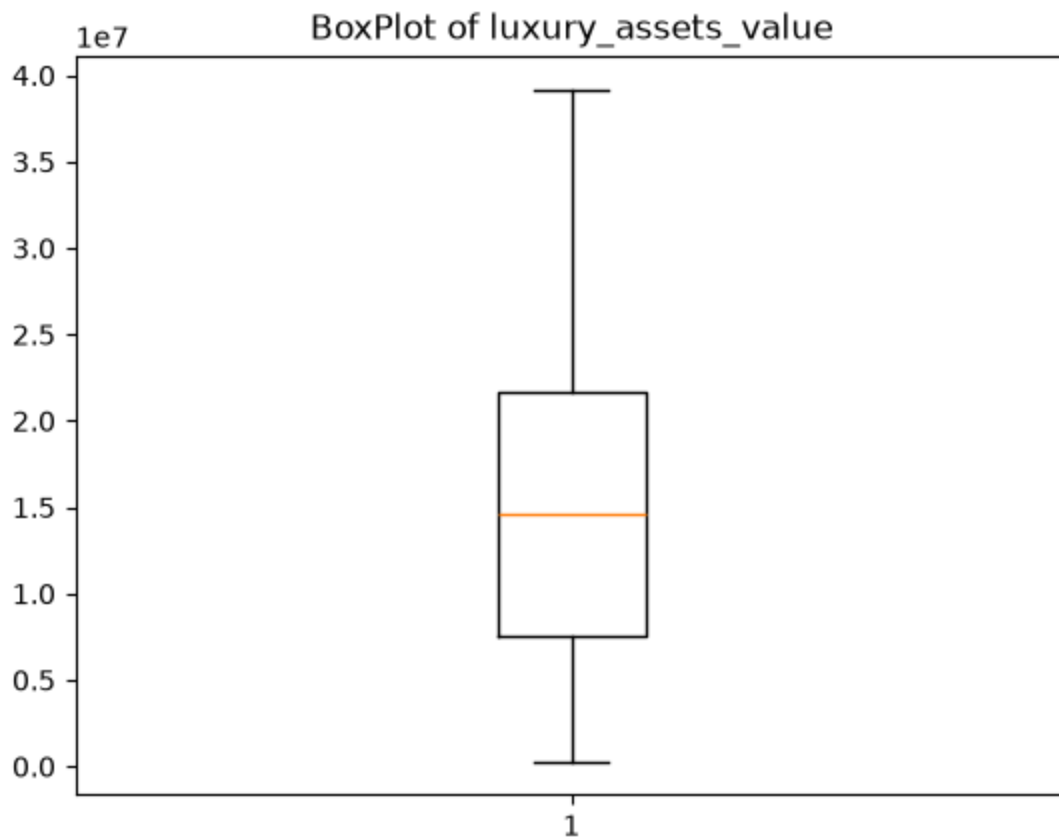


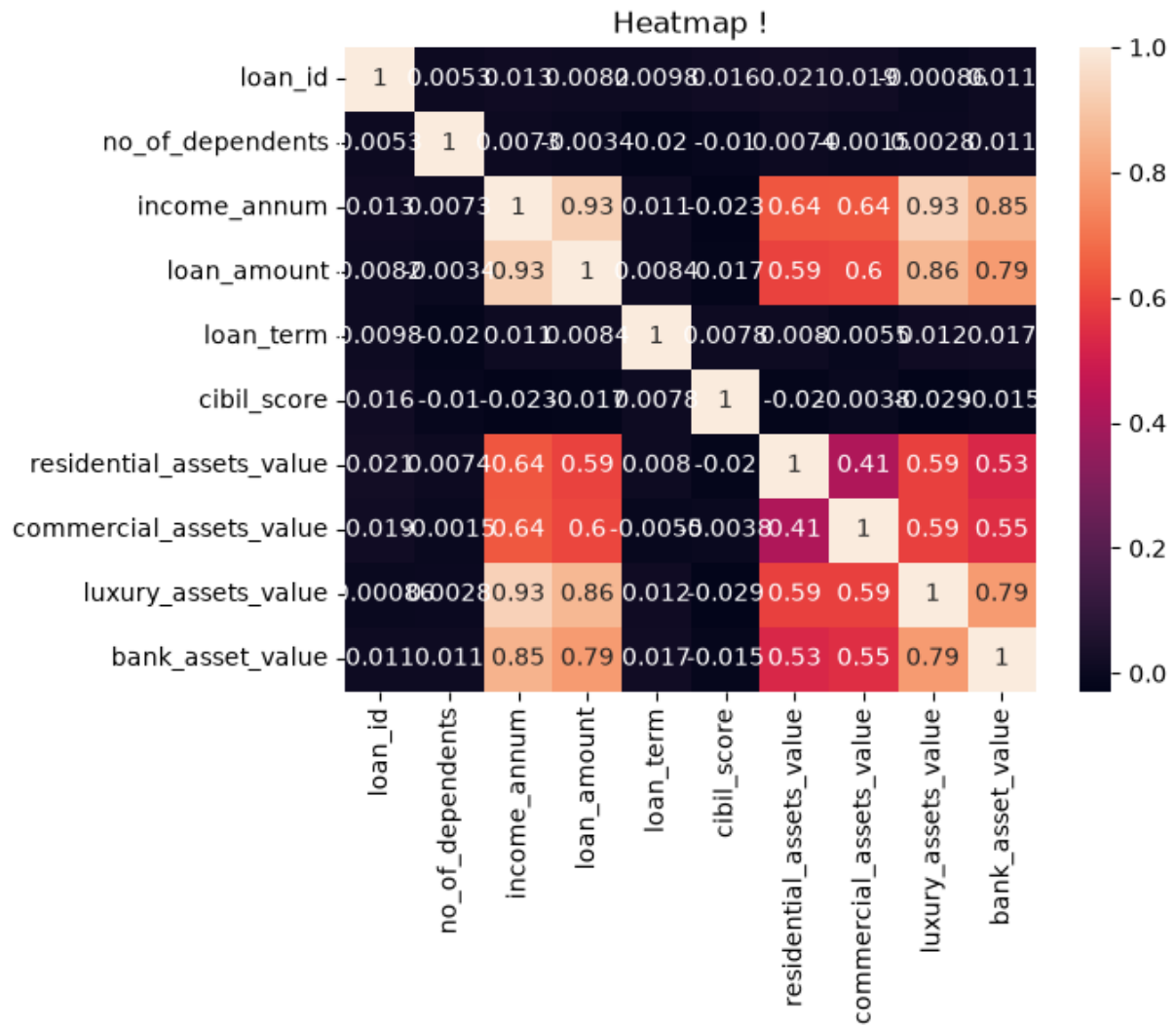


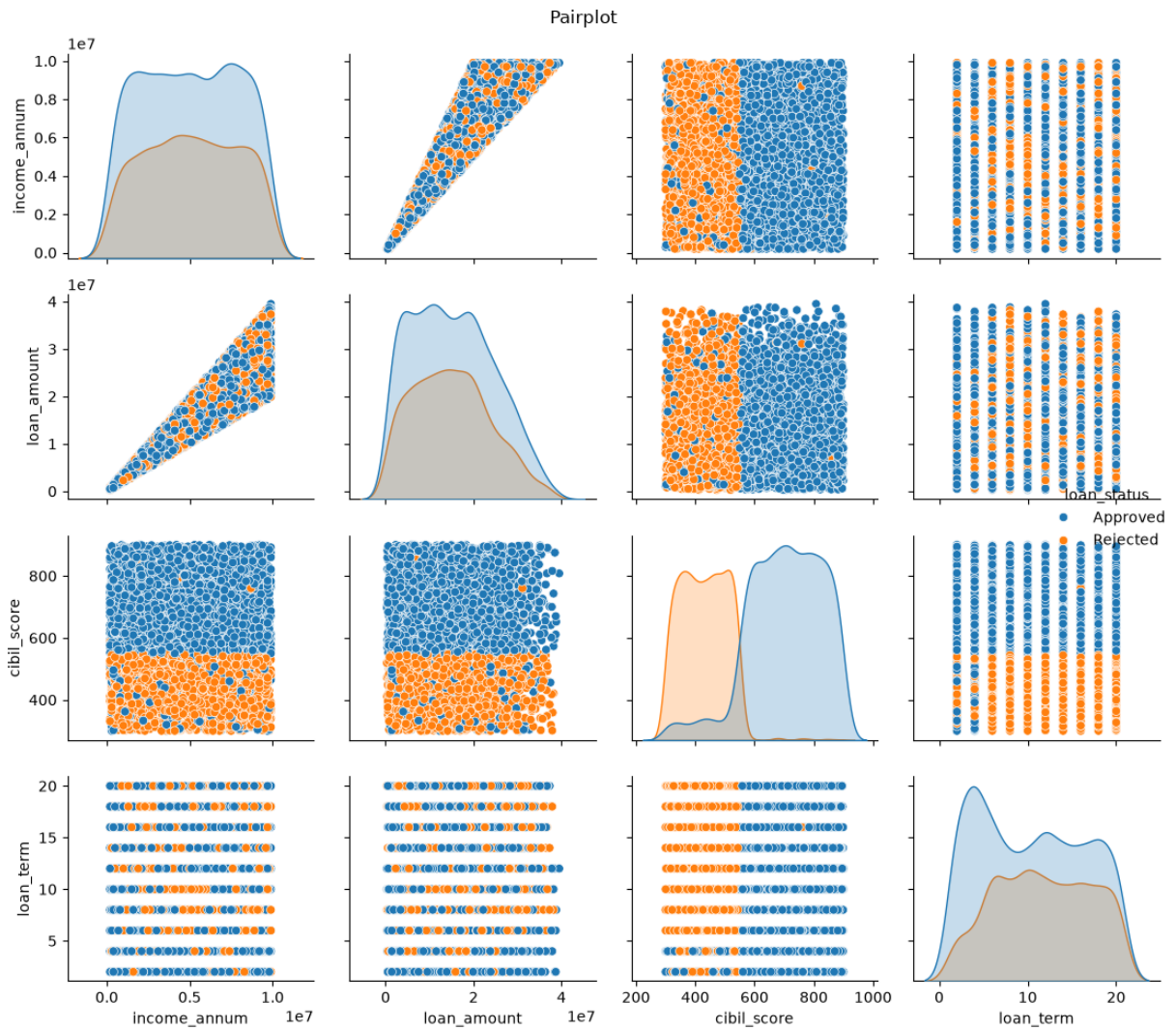












Observation - Loan Status Distribution

- The dataset contains both approved and rejected loan applications.
- Understanding this distribution helps identify whether the dataset is balanced or imbalanced before model training.
- We can see Aquired is higher in distribution!

Observation - Education Distribution

- Applicants are divided into Graduate and Not Graduate categories.
- Both graduate and not graduate are equal and hence balanced

Observation - Self Employed Distribution

- Applicants are categorized as self-employed or non-self-employed.
- It is balanced!

<h2 id="Observation---Histogram">Observation - Histogram</h2>

- Income_annum, loan_term, and cibil_score are all fairly **uniform**, with no distinct peak.
- Loan_amount and all four asset-value columns are **right-skewed**, concentrated at lower values with a long tail toward the maximum.

<h2 id="Observation---Box-Plot">Observation - Box Plot</h2>

- Residential_assets_value, commercial_assets_value, and bank_asset_value all show **upper outliers**.
- Income_annum, loan_amount, loan_term, and cibil_score show **no outliers**.
- We can see (residential_asset_value , commercial_asset_value and bank asset value having outliers

<h2 id="Observation---Correlation-Heatmap">Observation - Correlation Heatmap</h2>

- Income_annum and loan_amount show a **strong positive correlation (0.93)**.
- Income also correlates strongly with luxury_assets_value (**0.93**) and bank_asset_value (**0.85**).
- Cibil_score shows **very poor correlation** with every other feature (**-0.02 to 0.03**), including loan_amount.
- Asset-value features show moderate correlation with one another (**0.41–0.79**).

<h2 id="Observation---Pair-Plot">Observation - Pair Plot</h2>

- Cibil_score is the only feature where approved and rejected applications form **visibly separated distributions** — rejected applications skew lower, approved skew higher.
- Income_annum, loan_amount, and loan_term show **heavy overlap** between the two classes with no clear separation.
- This suggests cibil_score, despite its poor correlation with other numerical features, is the **strongest visual predictor of loan_status**.

In []:

```
In [18]: def perform_eda(
          filepath,
          target_column=None,
          categorical_columns=None,
          pairplot_columns=None,
          column_names=None,
          separator=","):

    df=load_dataset(filepath,column_names=column_names,separator=separator)
    print("\nLoading dataset completed\n")

    dataset_info(df)
```

```
missing_values(df)
duplicate_info(df)
summary_statistics(df)

if categorical_columns is not None:
    for col in categorical_columns:
        class_distribution(df,col)

plot_histogram(df)

plot_boxplot(df)

plot_heatmap(df)

if pairplot_columns is not None:
    if tatget_column is not None:
        plot_scatterplot(df[pairplot_columns],target_column)
    else:
        plot_scatterplot(df[pairplot_columns])

return df
```

```
In [16]: # df=perform_eda("emailSpam/emails.csv")
```

<h1 id="3.-Predicting-Diabetes">**3. Predicting Diabetes**</h1>

- Classification of whether a person has diabetes

```
In [19]: df=perform_eda("Diabetes_Prediction/diabetes_prediction_dataset.csv")
```

Loading dataset completed

 DATASET INFORMATION

Shape: (100000, 9)

Columns:

['gender', 'age', 'hypertension', 'heart_disease', 'smoking_history', 'bmi', 'HbA1c_level', 'blood_glucose_level', 'diabetes']

Data Types:

```
gender          str
age             float64
hypertension    int64
heart_disease   int64
smoking_history str
bmi             float64
HbA1c_level     float64
blood_glucose_level int64
diabetes        int64
dtype: object
```

Memory information

<class 'pandas.DataFrame'>

RangeIndex: 100000 entries, 0 to 99999

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	gender	100000 non-null	str
1	age	100000 non-null	float64
2	hypertension	100000 non-null	int64
3	heart_disease	100000 non-null	int64
4	smoking_history	100000 non-null	str
5	bmi	100000 non-null	float64
6	HbA1c_level	100000 non-null	float64
7	blood_glucose_level	100000 non-null	int64
8	diabetes	100000 non-null	int64

dtypes: float64(3), int64(4), str(2)

memory usage: 6.9 MB

First 5 elements

	gender	age	hypertension	heart_disease	smoking_history	bmi	\
0	Female	80.0	0	1	never	25.19	
1	Female	54.0	0	0	No Info	27.32	
2	Male	28.0	0	0	never	27.32	
3	Female	36.0	0	0	current	23.45	
4	Male	76.0	1	1	current	20.14	

	HbA1c_level	blood_glucose_level	diabetes
0	6.6	140	0
1	6.6	80	0
2	5.7	158	0
3	5.0	155	0
4	4.8	155	0

missing percent

```

gender          0      0.0
age             0      0.0
hypertension    0      0.0
heart_disease   0      0.0
smoking_history 0      0.0
bmi             0      0.0
HbA1c_level     0      0.0
blood_glucose_level 0    0.0
diabetes        0      0.0

```

There is no missing value!

The duplicate rows are : 3854

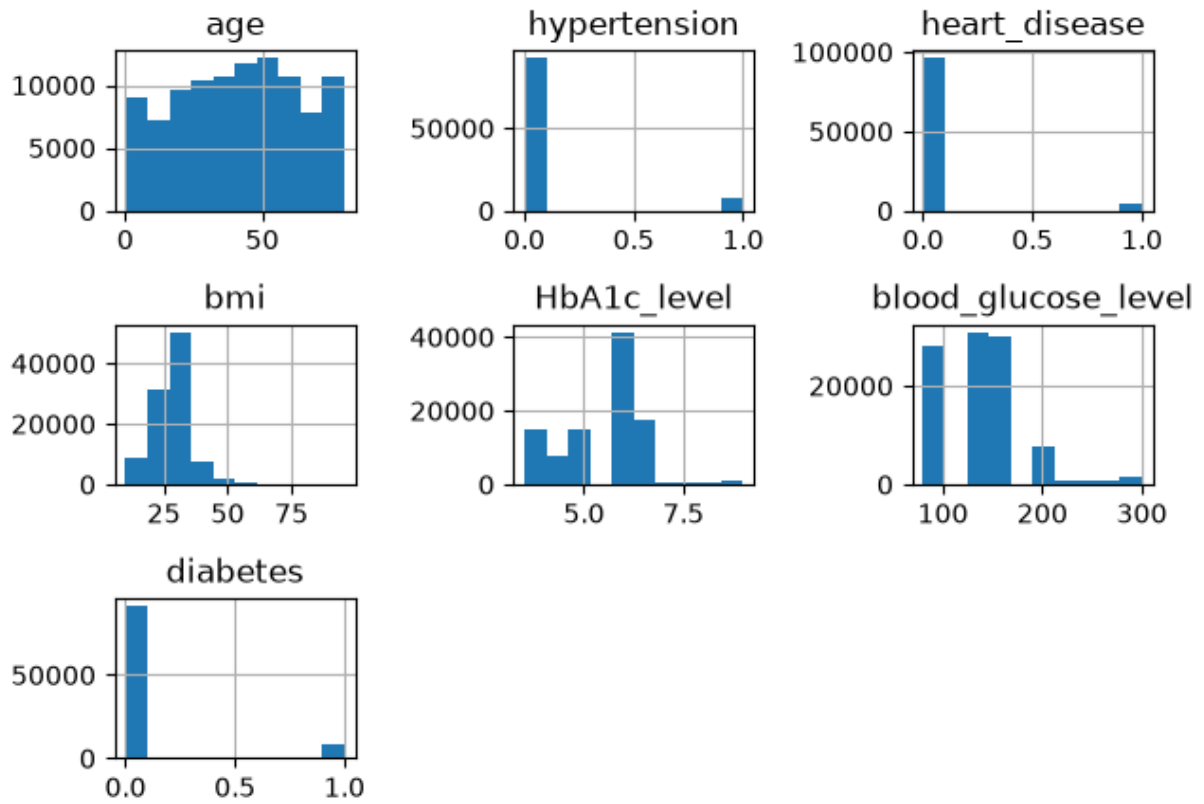
Duplicates are detected

Summary statistics

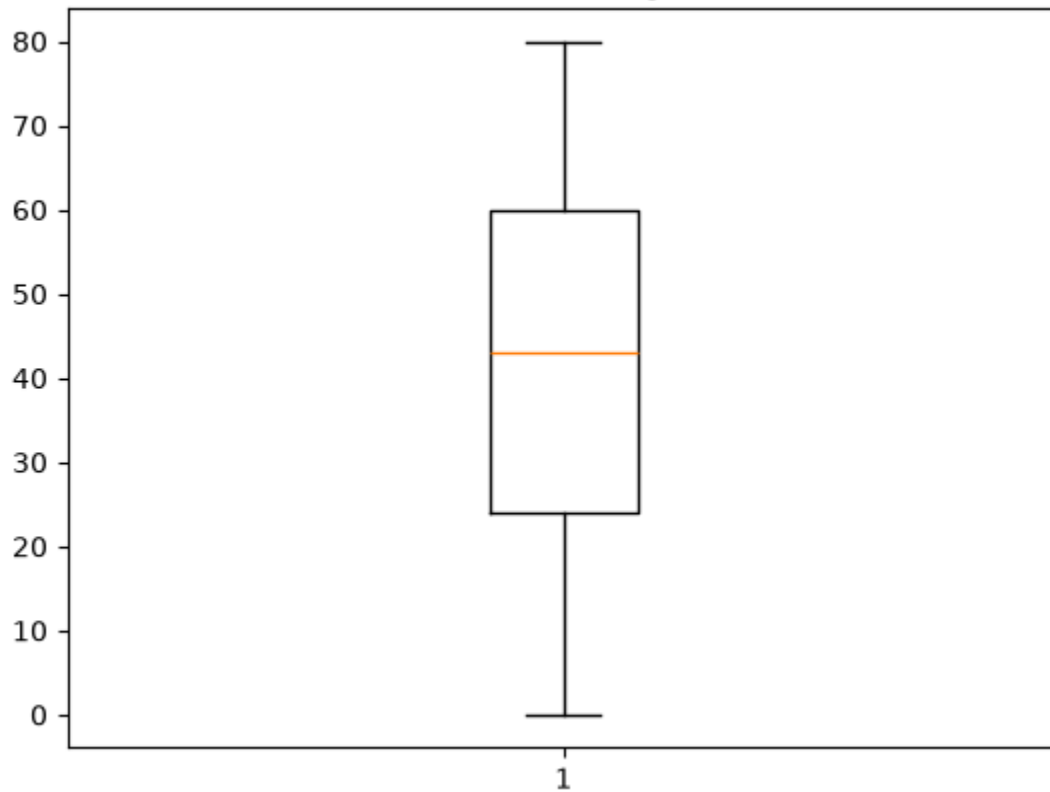
	age	hypertension	heart_disease	bmi \
count	100000.000000	100000.000000	100000.000000	100000.000000
mean	41.885856	0.07485	0.039420	27.320767
std	22.516840	0.26315	0.194593	6.636783
min	0.080000	0.00000	0.000000	10.010000
25%	24.000000	0.00000	0.000000	23.630000
50%	43.000000	0.00000	0.000000	27.320000
75%	60.000000	0.00000	0.000000	29.580000
max	80.000000	1.00000	1.000000	95.690000

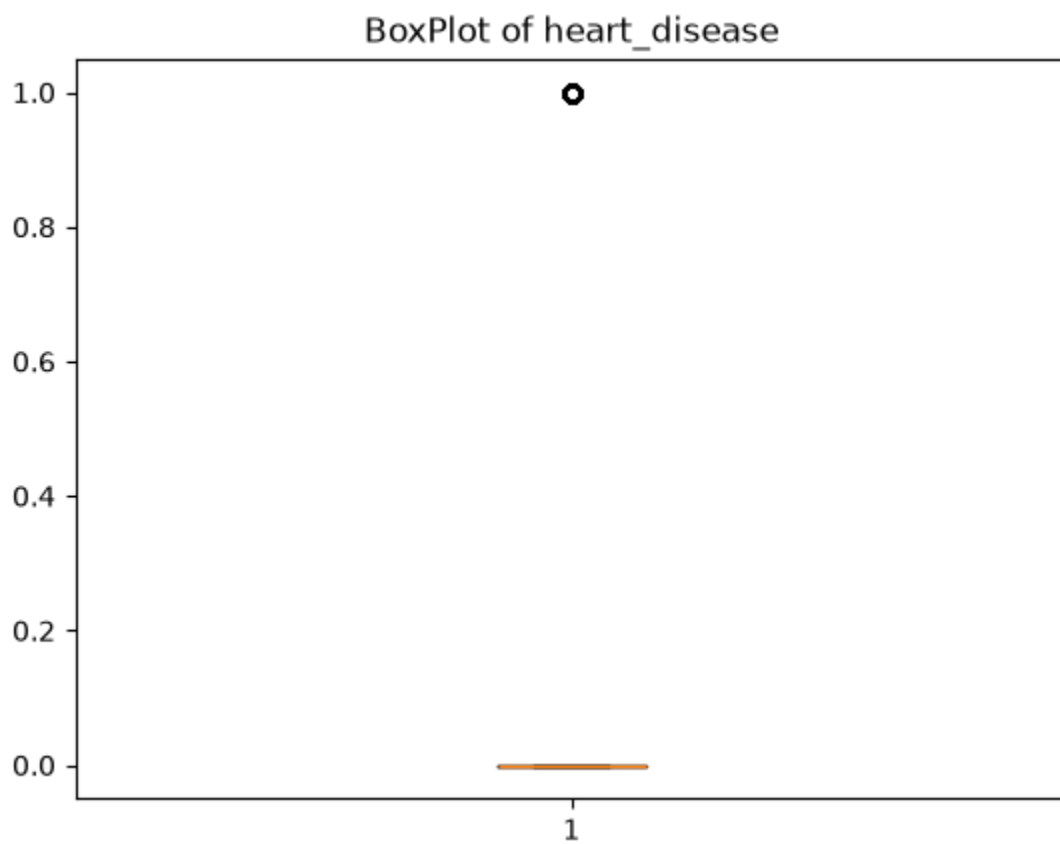
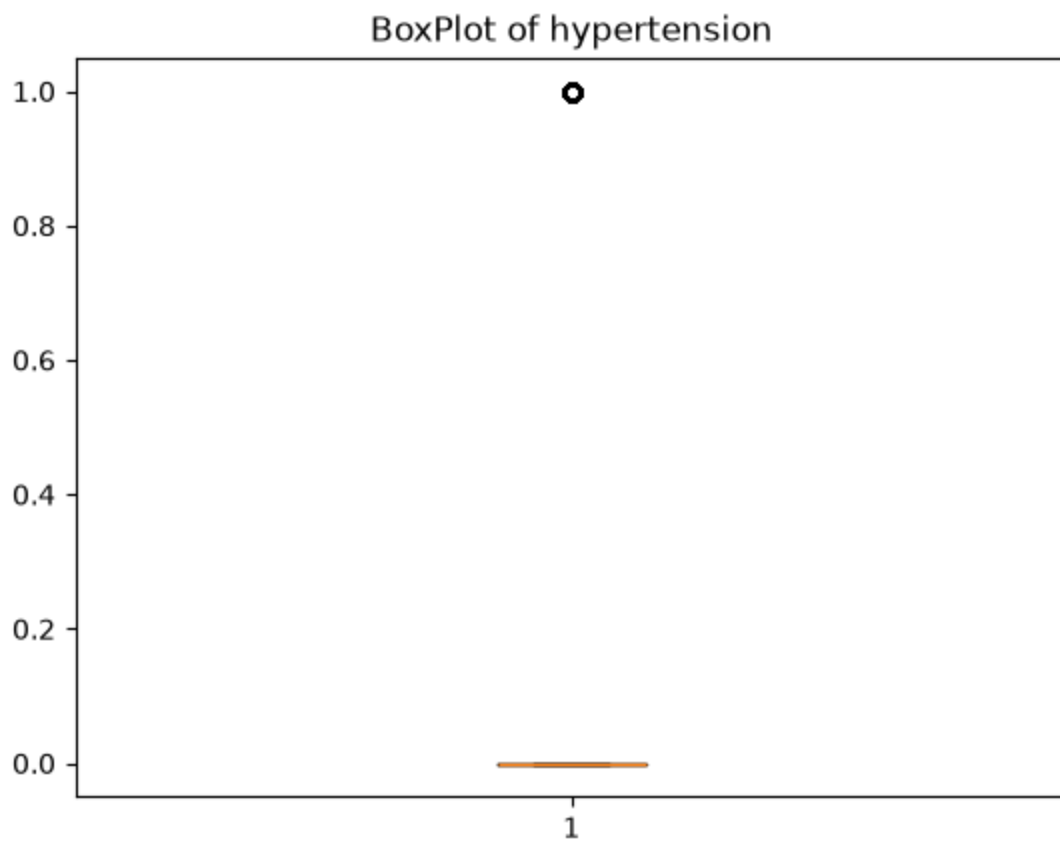
	HbA1c_level	blood_glucose_level	diabetes
count	100000.000000	100000.000000	100000.000000
mean	5.527507	138.058060	0.085000
std	1.070672	40.708136	0.278883
min	3.500000	80.000000	0.000000
25%	4.800000	100.000000	0.000000
50%	5.800000	140.000000	0.000000
75%	6.200000	159.000000	0.000000
max	9.000000	300.000000	1.000000

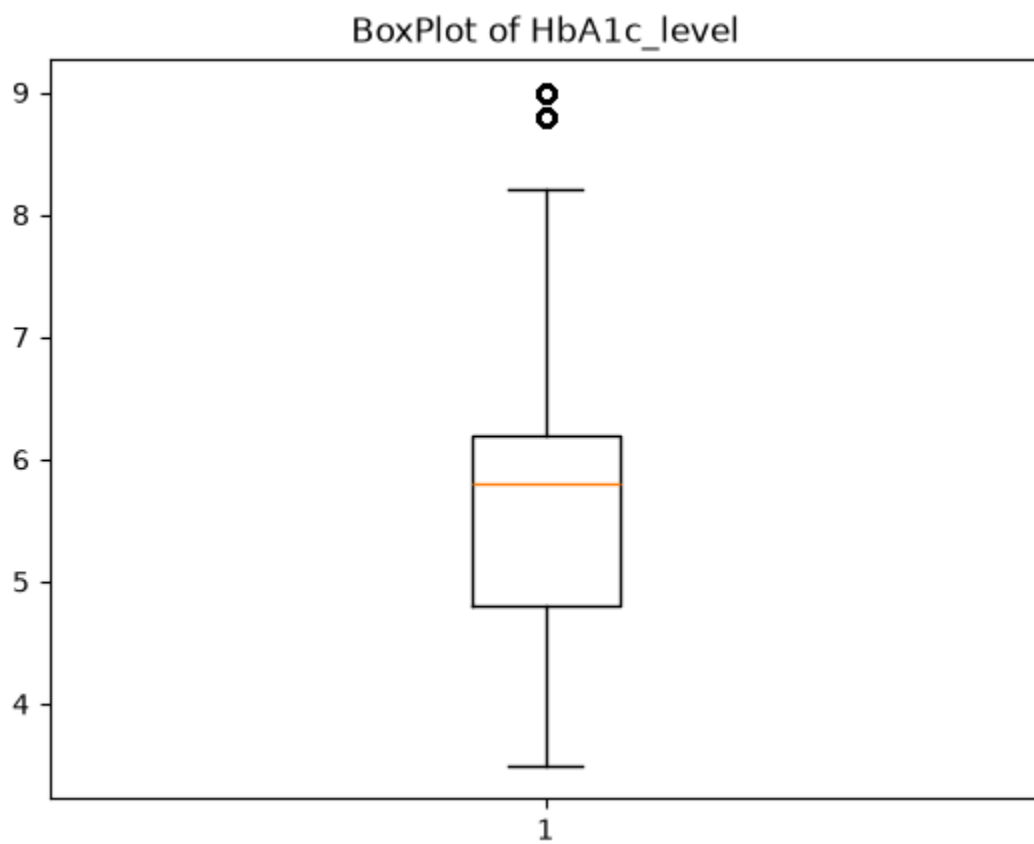
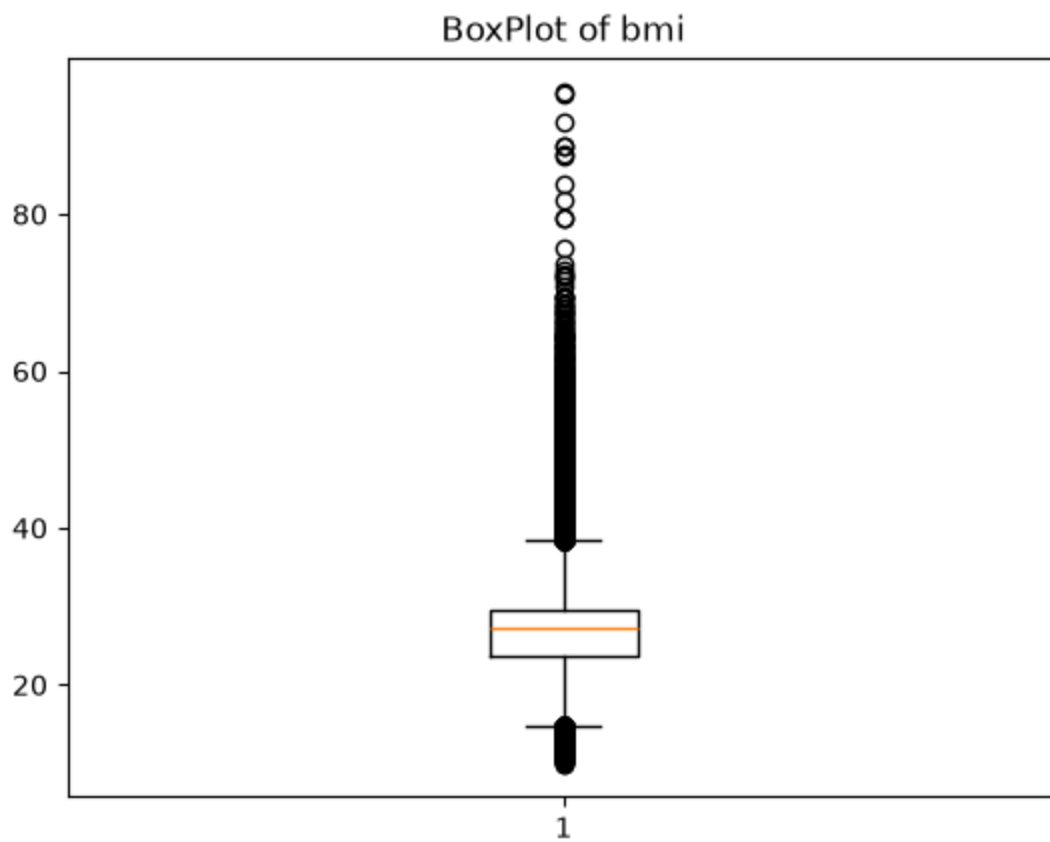
Histogram of numerical features

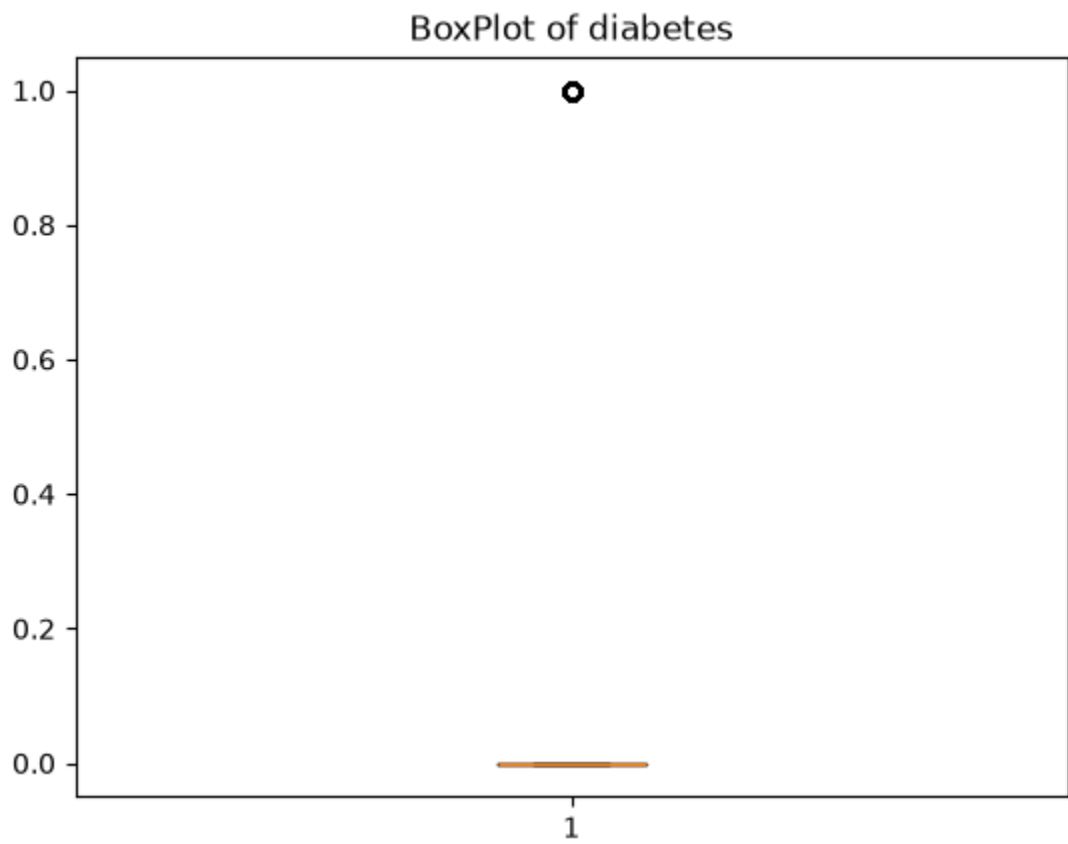
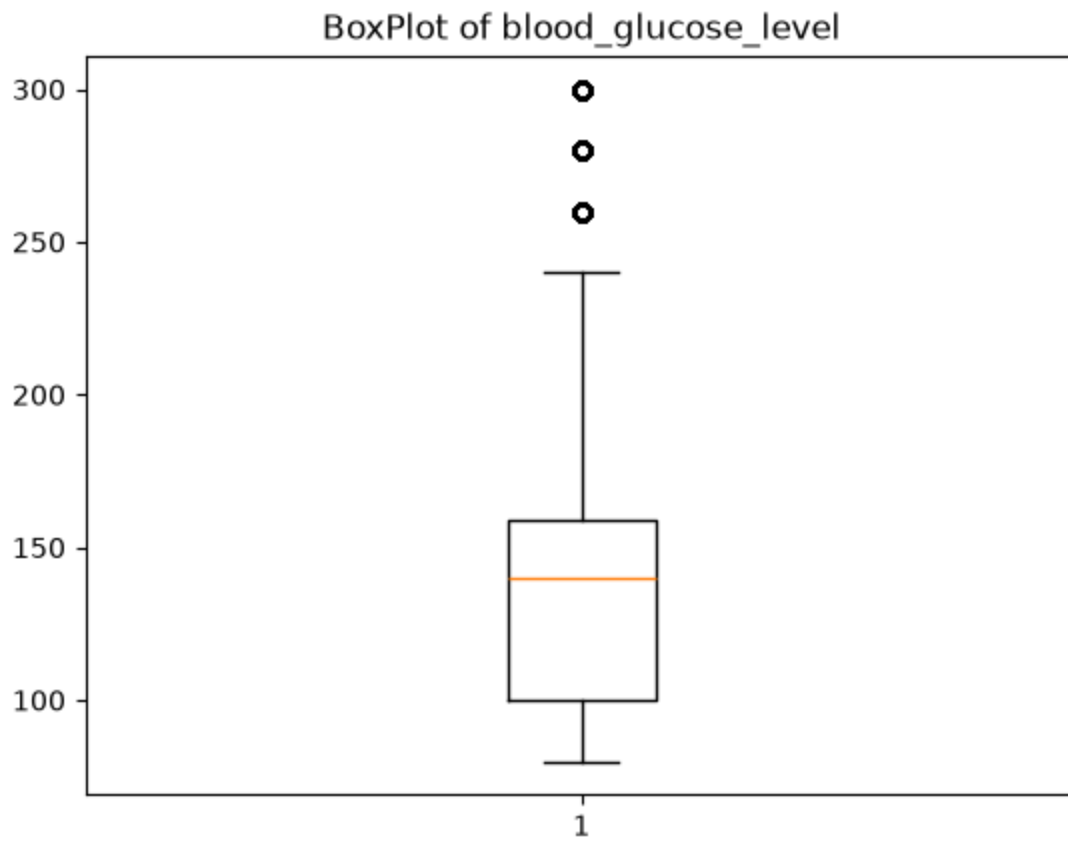


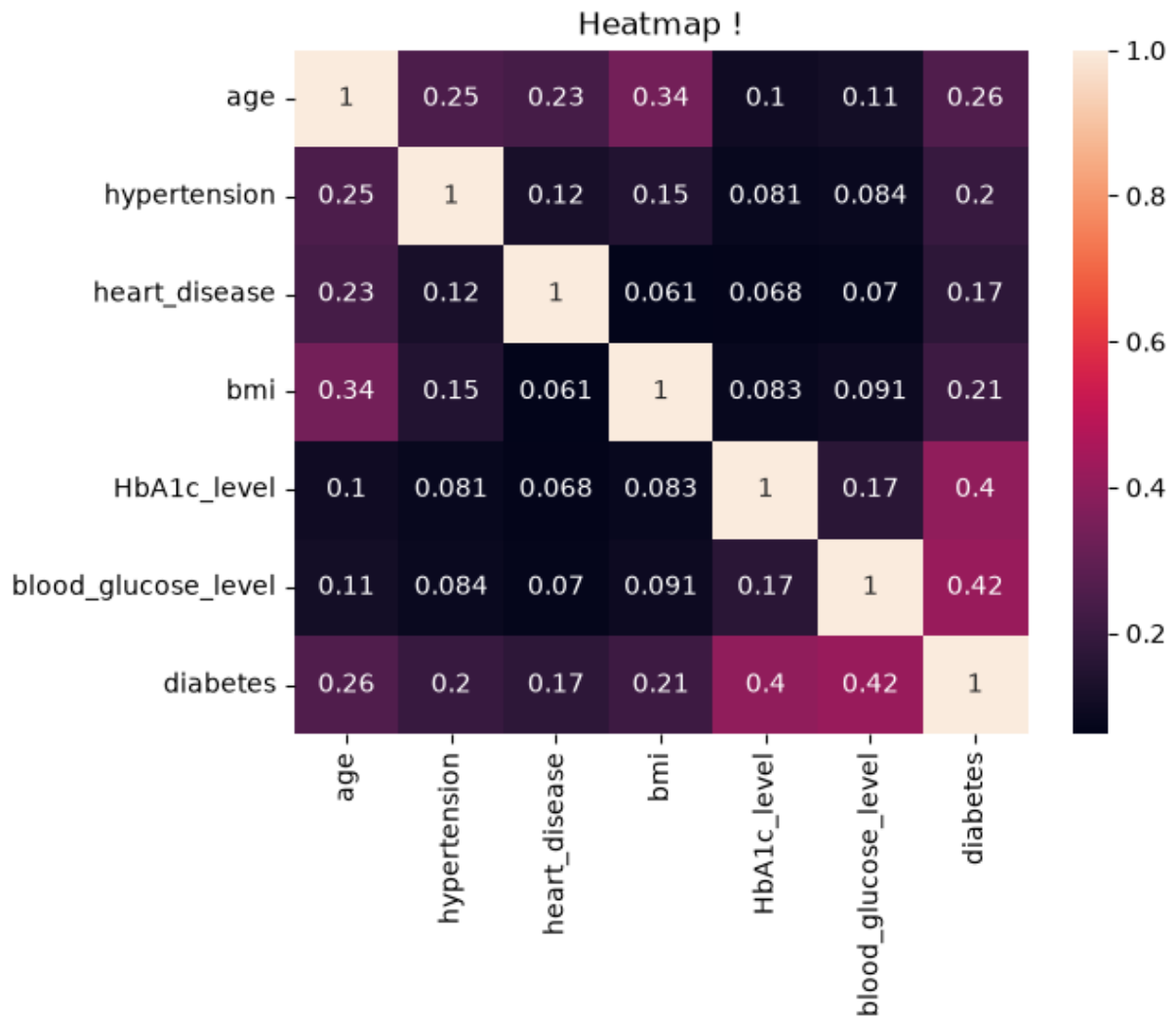
BoxPlot of age











Observation - Dataset Information

- The dataset contains **100,000 records** and **9 features**.
- There are **8 input features** and **1 target feature** (diabetes), along with categorical attributes such as gender and smoking history.
- The dataset consists of both numerical and categorical variables describing patient health and lifestyle.
- The dataset is suitable for a supervised binary classification task.

Observation - Missing Values

- No missing values are present in any of the dataset columns.
- Since the dataset is complete, no missing value treatment is required.

Observation - Duplicate Values

- A total of **3,854 duplicate records** were detected in the dataset.
- Duplicate records may introduce redundancy and bias during model training.
- Duplicates need to be dealt with in preprocessing stage

<h2 id="Observation---Summary-Statistics">Observation - Summary Statistics</h2>

- Patient ages range from **0.08 years to 80 years**, indicating that the dataset includes individuals from infancy to old age.
- BMI values range from **10.01 to 95.69**, showing considerable variation in body mass among patients.
- HbA1c levels vary between **3.5 and 9.0**, representing different levels of long-term blood glucose control.
- Blood glucose levels range from **80 to 300 mg/dL**, covering both normal and diabetic conditions.
- The diabetes target variable is binary, with values **0 (Non-Diabetic)** and **1 (Diabetic)**.

<h2 id="Observation---Histogram">Observation - Histogram</h2>

- Age is distributed across a wide range, with a higher concentration of patients between approximately 20 and 60 years.
- Hypertension and heart disease are highly imbalanced, as most patients belong to class 0 (absence of the condition).
- BMI values are concentrated around the mid-25s, with relatively fewer patients having very low or very high BMI.
- HbA1c levels show a prominent peak around 6.0, with smaller concentrations between approximately 3.5 and 5.0.
- Very high HbA1c values (above about 7) occur much less frequently.
- Blood glucose levels are mainly concentrated between 80 and 160, with fewer observations at higher levels.
- The diabetes target variable is highly imbalanced, with non-diabetic patients (0) significantly outnumbering diabetic patients (1).

<h2 id="Observation---Box-Plot">Observation - Box Plot</h2>

- Age does not exhibit significant outliers and is spread across the entire age range.
- BMI contains numerous upper outliers, indicating the presence of patients with unusually high BMI values.
- HbA1c level shows a few high-value outliers above the upper whisker.
- Blood glucose level also contains several upper outliers, with values extending close to 300.
- Binary features such as hypertension, heart disease, and diabetes show limited variation due to their binary nature.

<h2 id="Observation---Heatmap">Observation - Heatmap</h2>

- Most numerical features exhibit weak to moderate positive correlations.
- Blood glucose level has the highest correlation with diabetes (0.42).
- HbA1c level also shows a moderate positive correlation with diabetes (0.40).

- Age and BMI have weak positive correlations with diabetes (0.26 and 0.21 respectively).
- No pair of features exhibits very strong multicollinearity, indicating that the numerical features provide distinct information.

```
In [22]: def perform_eda(
    filepath,
    target_column=None,
    categorical_columns=None,
    pairplot_columns=None,
    column_names=None,
    separator=" ",

    show_histogram=True,
    show_boxplot=True,
    show_heatmap=True,
    show_pairplot=True
):
    # -----
    # Load Dataset
    # -----
    df = load_dataset(
        filepath,
        column_names=column_names,
        separator=separator
    )

    print("\nLoading dataset completed\n")

    # -----
    # Basic Information
    # -----
    dataset_info(df)
    missing_values(df)
    duplicate_info(df)
    summary_statistics(df)

    # -----
    # Class Distribution
    # -----
    if categorical_columns is not None:
        for col in categorical_columns:
            class_distribution(df, col)

    # -----
    # Visualizations
    # -----

    if show_histogram:
        plot_histogram(df)

    if show_boxplot:
        plot_boxplot(df)
```

```
if show_heatmap:
    plot_heatmap(df)

if show_pairplot:

    if pairplot_columns is not None:

        if target_column is not None:
            plot_scatterplot(
                df[pairplot_columns + [target_column]],
                target_column
            )

        else:
            plot_scatterplot(
                df[pairplot_columns]
            )

    else:
        print("\nPairplot skipped (No columns specified).")

return df
```

<h1 id="4.-handwritten-Character-recognition">4. handwritten Character recognition</h1>

```
In [23]: from sklearn.datasets import load_digits
import pandas as pd

digits = load_digits()
df= pd.DataFrame(digits.data)
df["target"]=digits.target

df.to_csv("digits.csv",index=False)
```

```
In [25]: df = perform_eda(
    "digits.csv",
    categorical_columns=["target"],

    show_histogram=False,
    show_boxplot=False,
    show_heatmap=False,
    show_pairplot=False
)
```

Loading dataset completed

 DATASET INFORMATION

Shape: (1797, 65)

Columns:

['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', '16', '17', '18', '19', '20', '21', '22', '23', '24', '25', '26', '27', '28', '29', '30', '31', '32', '33', '34', '35', '36', '37', '38', '39', '40', '41', '42', '43', '44', '45', '46', '47', '48', '49', '50', '51', '52', '53', '54', '55', '56', '57', '58', '59', '60', '61', '62', '63', 'target']

Data Types:

0 float64
 1 float64
 2 float64
 3 float64
 4 float64

...
 60 float64
 61 float64
 62 float64
 63 float64

target int64

Length: 65, dtype: object

Memory information

<class 'pandas.DataFrame'>

RangeIndex: 1797 entries, 0 to 1796

Data columns (total 65 columns):

#	Column	Non-Null Count	Dtype
0	0	1797 non-null	float64
1	1	1797 non-null	float64
2	2	1797 non-null	float64
3	3	1797 non-null	float64
4	4	1797 non-null	float64
5	5	1797 non-null	float64
6	6	1797 non-null	float64
7	7	1797 non-null	float64
8	8	1797 non-null	float64
9	9	1797 non-null	float64
10	10	1797 non-null	float64
11	11	1797 non-null	float64
12	12	1797 non-null	float64
13	13	1797 non-null	float64
14	14	1797 non-null	float64
15	15	1797 non-null	float64
16	16	1797 non-null	float64
17	17	1797 non-null	float64
18	18	1797 non-null	float64
19	19	1797 non-null	float64

```

20 20      1797 non-null float64
21 21      1797 non-null float64
22 22      1797 non-null float64
23 23      1797 non-null float64
24 24      1797 non-null float64
25 25      1797 non-null float64
26 26      1797 non-null float64
27 27      1797 non-null float64
28 28      1797 non-null float64
29 29      1797 non-null float64
30 30      1797 non-null float64
31 31      1797 non-null float64
32 32      1797 non-null float64
33 33      1797 non-null float64
34 34      1797 non-null float64
35 35      1797 non-null float64
36 36      1797 non-null float64
37 37      1797 non-null float64
38 38      1797 non-null float64
39 39      1797 non-null float64
40 40      1797 non-null float64
41 41      1797 non-null float64
42 42      1797 non-null float64
43 43      1797 non-null float64
44 44      1797 non-null float64
45 45      1797 non-null float64
46 46      1797 non-null float64
47 47      1797 non-null float64
48 48      1797 non-null float64
49 49      1797 non-null float64
50 50      1797 non-null float64
51 51      1797 non-null float64
52 52      1797 non-null float64
53 53      1797 non-null float64
54 54      1797 non-null float64
55 55      1797 non-null float64
56 56      1797 non-null float64
57 57      1797 non-null float64
58 58      1797 non-null float64
59 59      1797 non-null float64
60 60      1797 non-null float64
61 61      1797 non-null float64
62 62      1797 non-null float64
63 63      1797 non-null float64
64 target 1797 non-null int64

```

dtypes: float64(64), int64(1)

memory usage: 912.7 KB

First 5 elements

	0	1	2	3	4	5	6	7	8	9	...	55	56	57
\														
0	0.0	0.0	5.0	13.0	9.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0
1	0.0	0.0	0.0	12.0	13.0	5.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0
2	0.0	0.0	0.0	4.0	15.0	12.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0
3	0.0	0.0	7.0	15.0	13.0	1.0	0.0	0.0	0.0	8.0	...	0.0	0.0	0.0
4	0.0	0.0	0.0	1.0	11.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0

	58	59	60	61	62	63	target
0	6.0	13.0	10.0	0.0	0.0	0.0	0
1	0.0	11.0	16.0	10.0	0.0	0.0	1
2	0.0	3.0	11.0	16.0	9.0	0.0	2
3	7.0	13.0	13.0	9.0	0.0	0.0	3
4	0.0	2.0	16.0	4.0	0.0	0.0	4

```
[5 rows x 65 columns]
missing percent
0          0      0.0
1          0      0.0
2          0      0.0
3          0      0.0
4          0      0.0
...
60         0      0.0
61         0      0.0
62         0      0.0
63         0      0.0
target     0      0.0
```

```
[65 rows x 2 columns]
There is no missing value!
```

```
The duplicate rows are : 0
No duplicates detected!
```

```
Summary statistics
```

	0	1	2	3	4 \
count	1797.0	1797.000000	1797.000000	1797.000000	1797.000000
mean	0.0	0.303840	5.204786	11.835838	11.848080
std	0.0	0.907192	4.754826	4.248842	4.287388
min	0.0	0.000000	0.000000	0.000000	0.000000
25%	0.0	0.000000	1.000000	10.000000	10.000000
50%	0.0	0.000000	4.000000	13.000000	13.000000
75%	0.0	0.000000	9.000000	15.000000	15.000000
max	0.0	8.000000	16.000000	16.000000	16.000000

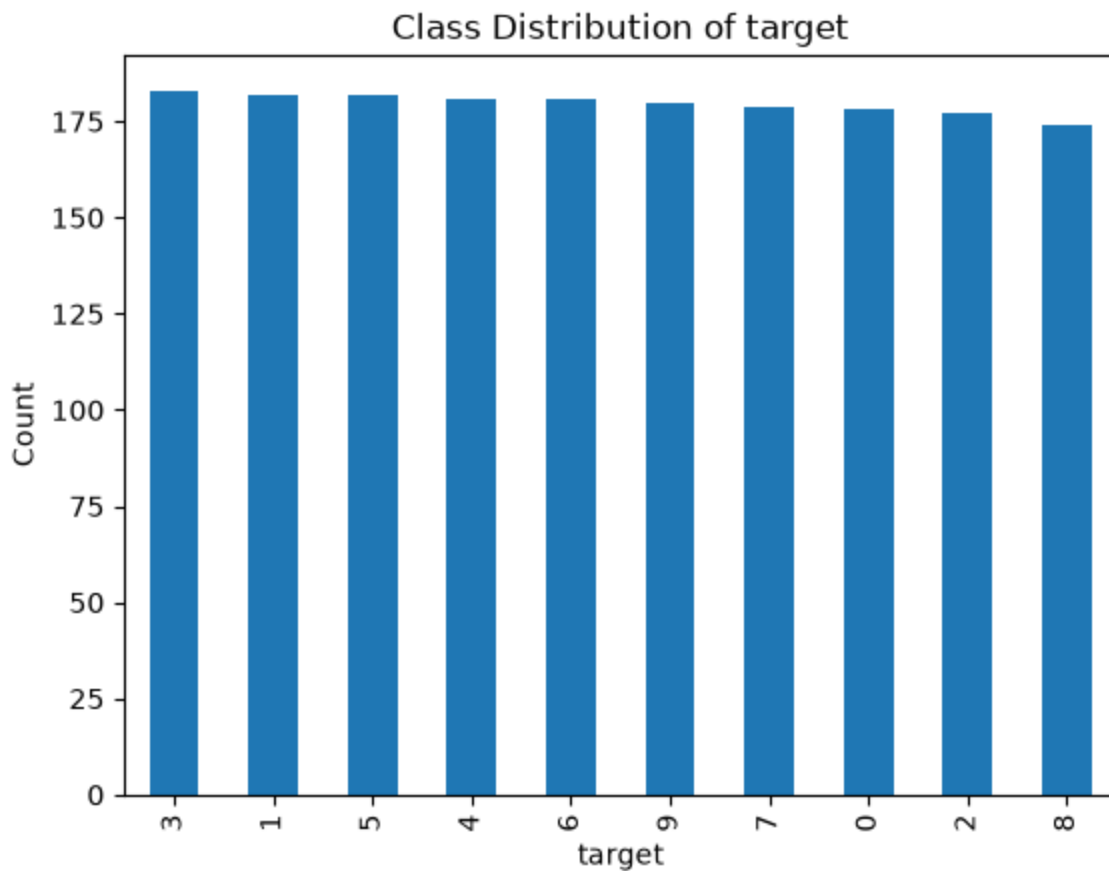
	5	6	7	8	9 ...
\count	1797.000000	1797.000000	1797.000000	1797.000000	1797.000000 ...
mean	5.781859	1.362270	0.129661	0.005565	1.993879 ...
std	5.666418	3.325775	1.037383	0.094222	3.196160 ...
min	0.000000	0.000000	0.000000	0.000000	0.000000 ...
25%	0.000000	0.000000	0.000000	0.000000	0.000000 ...
50%	4.000000	0.000000	0.000000	0.000000	0.000000 ...
75%	11.000000	0.000000	0.000000	0.000000	3.000000 ...
max	16.000000	16.000000	15.000000	2.000000	16.000000 ...

	55	56	57	58	59 \
count	1797.000000	1797.000000	1797.000000	1797.000000	1797.000000
mean	0.206455	0.000556	0.279354	5.557596	12.089037
std	0.984401	0.023590	0.934302	5.103019	4.374694
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	1.000000	11.000000
50%	0.000000	0.000000	0.000000	4.000000	13.000000

75%	0.000000	0.000000	0.000000	10.000000	16.000000
max	13.000000	1.000000	9.000000	16.000000	16.000000
	60	61	62	63	target
count	1797.000000	1797.000000	1797.000000	1797.000000	1797.000000
mean	11.809126	6.764051	2.067891	0.364496	4.490818
std	4.933947	5.900623	4.090548	1.860122	2.865304
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	10.000000	0.000000	0.000000	0.000000	2.000000
50%	14.000000	6.000000	0.000000	0.000000	4.000000
75%	16.000000	12.000000	2.000000	0.000000	7.000000
max	16.000000	16.000000	16.000000	16.000000	9.000000

[8 rows x 65 columns]

Class Distribution



<h1 id="5.-Classification-Email-Spam">5. Classification Email Spam</h1>

- d

```
In [27]: perform_eda(
    "emailSpam/emails.csv",
    categorical_columns=["Prediction"],

    show_histogram=False,
    show_boxplot=False,
    show_heatmap=False,
    show_pairplot=False
```

)

Loading dataset completed

DATASET INFORMATION

Shape: (5172, 3002)

Columns:

['Email No.', 'the', 'to', 'ect', 'and', 'for', 'of', 'a', 'you', 'hou', 'in',
, 'on', 'is', 'this', 'enron', 'i', 'be', 'that', 'will', 'have', 'with', '
your', 'at', 'we', 's', 'are', 'it', 'by', 'com', 'as', 'from', 'gas', 'or',
'not', 'me', 'deal', 'if', 'meter', 'hpl', 'please', 're', 'e', 'any', 'our',
, 'corp', 'can', 'd', 'all', 'has', 'was', 'know', 'need', 'an', 'forwarded',
, 'new', 't', 'may', 'up', 'j', 'mmbtu', 'should', 'do', 'am', 'get', 'out',
'see', 'no', 'there', 'price', 'daren', 'but', 'been', 'company', 'l', 'thes
e', 'let', 'so', 'would', 'm', 'into', 'xls', 'farmer', 'attached', 'us', 'i
nformation', 'they', 'message', 'day', 'time', 'my', 'one', 'what', 'only',
'http', 'th', 'volume', 'mail', 'contract', 'which', 'month', 'more', 'rober
t', 'sitara', 'about', 'texas', 'nom', 'energy', 'pec', 'questions', 'www',
'deals', 'volumes', 'pm', 'ena', 'now', 'their', 'file', 'some', 'email', 'j
ust', 'also', 'call', 'change', 'other', 'here', 'like', 'b', 'flow', 'net',
'following', 'p', 'production', 'when', 'over', 'back', 'want', 'original',
'them', 'below', 'o', 'ticket', 'c', 'he', 'could', 'make', 'inc', 'report',
'march', 'contact', 'were', 'days', 'list', 'nomination', 'system', 'who', '
april', 'number', 'sale', 'don', 'its', 'first', 'thanks', 'business', 'help
, 'per', 'through', 'july', 'forward', 'font', 'free', 'daily', 'use', 'ord
er', 'today', 'r', 'had', 'fw', 'set', 'plant', 'statements', 'go', 'gary',
'oil', 'line', 'sales', 'w', 'effective', 'well', 'tenaska', 'take', 'june',
'x', 'within', 'nbsp', 'she', 'how', 'north', 'america', 'being', 'under', '
next', 'week', 'than', 'january', 'last', 'two', 'service', 'purchase', 'nam
e', 'less', 'height', 'off', 'agreement', 'k', 'work', 'tap', 'group', 'year
, 'based', 'transport', 'after', 'think', 'made', 'each', 'available', 'cha
nges', 'due', 'f', 'h', 'services', 'smith', 'send', 'management', 'stock',
'sent', 'll', 'co', 'office', 'needs', 'cotten', 'did', 'actuals', 'u', 'mon
ey', 'before', 'looking', 'then', 'pills', 'online', 'request', 'look', 'des
k', 'ami', 'his', 'same', 'george', 'chokshi', 'point', 'delivery', 'friday',
, 'does', 'size', 'august', 'product', 'pat', 'width', 'iv', 'noms', 'addres
s', 'above', 'sure', 'give', 'october', 'future', 'find', 'market', 'n', 'ma
ry', 'vance', 'melissa', 'said', 'internet', 'still', 'account', 'those', 'd
own', 'link', 'hsc', 'rate', 'people', 'pipeline', 'best', 'actual', 'very',
'end', 'home', 'houston', 'tu', 'high', 'her', 'team', 'products', 'many', '
currently', 'spot', 'receive', 'good', 'such', 'going', 'process', 'feb', 'm
onday', 'info', 'david', 'lloyd', 'again', 'both', 'click', 'subject', 'jack
ie', 'december', 'total', 'na', 'lisa', 've', 'september', 'hours', 'until',
'resources', 'because', 'aol', 'february', 'where', 'g', 'investment', 'issu
e', 'duke', 'since', 'pay', 'show', 'way', 'global', 'computron', 'further',
'most', 'place', 'offer', 'natural', 'activity', 'easttrans', 'graves', 'righ
t', 'prices', 'date', 'john', 'utilities', 'november', 'clynes', 'jan', 'sec
urities', 'meeting', 'susan', 'hplc', 'julie', 'able', 'received', 'align',
'term', 'id', 'revised', 'thursday', 'pg', 'fee', 'hplno', 'trading', 'addit
ional', 'site', 'txu', 'data', 'wellhead', 'reply', 'taylor', 'news', 'unify',
, 'michael', 'provide', 'note', 'much', 'access', 'lannou', 'every', 'betwe
en', 'keep', 'tuesday', 'review', 'great', 'tom', 'put', 'done', 'long', 'sa
ve', 'section', 'must', 'v', 'part', 'nd', 'million', 'check', 'trade', 'bob',
, 'created', 'steve', 'prior', 'copy', 'continue', 'numbers', 'via', 'world

, 'demand', 'hanks', 'contracts', 'phone', 'transaction', 'customer', 'possible', 'pefs', 'meyers', 'months', 'special', 'without', 'used', 'regarding', 'software', 'howard', 'support', 'buy', 'young', 'meters', 'thru', 'believe', 'gcs', 'cec', 'entered', 'control', 'dec', 'face', 'create', 'weissman', 'st', 'color', 'come', 'supply', 'brian', 'hplo', 'own', 'correct', 'customers', 'web', 'allocation', 'soon', 'using', 'development', 'mark', 'low', 'power', 'problem', 'once', 'however', 'tickets', 'border', 'performance', 'manager', 'rates', 'center', 'companies', 'risk', 'details', 'needed', 'international', 'field', 'even', 'someone', 'doc', 'fuel', 'lee', 'paid', 'while', 'start', 'index', 'include', 'nominations', 'act', 'pricing', 'scheduled', 'gathering', 'type', 'href', 'during', 'aimee', 'anything', 'feel', 'fuels', 'getting', 'advice', 'why', 'increase', 'path', 'sell', 'works', 'issues', 'three', 'enronxgate', 'camp', 'either', 'form', 'security', 'interest', 'financial', 'family', 'xp', 'plan', 'current', 'top', 'another', 'src', 'spreadsheet', 'allen', 'wednesday', 'read', 'him', 'working', 'wynne', 'add', 'deliveries', 'buyback', 'allocated', 'firm', 'james', 'marketing', 'tx', 'results', 'got', 'stocks', 'calpine', 'might', 'operations', 'position', 'logistics', 'fax', 'cost', 'party', 'zero', 'pops', 'old', 'pt', 'scheduling', 'flowed', 'dollars', 'update', 'gco', 'katy', 'including', 'follow', 'yahoo', 'already', 'suite', 'error', 'past', 'page', 'stop', 'changed', 'book', 'program', 'few', 'better', 'operating', 'equistar', 'move', 'cotton', 'aep', 'y', 'state', 'ees', 'rita', 'provided', 'employees', 'period', 'morning', 'cd', 'hotmail', 'entex', 'swing', 'real', 'exchange', 'tomorrow', 'lst', 'counterparty', 'parker', 'person', 'follows', 'valid', 'visit', 'little', 'professional', 'quality', 'confirm', 'something', 'megan', 'brenda', 'around', 'windows', 'im', 'storage', 'accounting', 'called', 'ranch', 'tax', 'problems', 'case', 'teco', 'fact', 'always', 'too', 'unsubscribe', 'amount', 'coastal', 'never', 'rodriguez', 'love', 'acton', 'shut', 'pipe', 'project', 'hope', 'limited', 'invoice', 'credit', 'full', 'survey', 'ray', 'carlos', 'anyone', 'wanted', 'yet', 'ic', 'scott', 'years', 'charlie', 'soft', 'notice', 'advise', 'addition', 'donald', 'lsk', 'wish', 'katherine', 'website', 'hplnl', 'schumack', 'prescription', 'cover', 'shares', 'cash', 'imbalance', 'united', 'handle', 'big', 'everyone', 'style', 'clear', 'producer', 'weekend', 'city', 'requested', 'stone', 'left', 'payment', 'mobil', 'shows', 'small', 'confirmed', 'technology', 'meet', 'extend', 'life', 'intended', 'sherlyn', 'schedule', 'else', 'letter', 'box', 'bill', 'richard', 'lamphier', 'complete', 'ever', 'release', 'newsletter', 'anita', 'clem', 'having', 'herod', 'beginning', 'papayoti', 'try', 'mike', 'enter', 'estimates', 'location', 'cut', 'question', 'things', 'personal', 'feedback', 'cialis', 'found', 'area', 'dow', 'terms', 'central', 'necessary', 'man', 'run', 'reason', 'third', 'midcon', 'charge', 'president', 'de', 'listed', 'meds', 'thomas', 'thought', 'capital', 'added', 'ask', 'weeks', 'investing', 'commercial', 'star', 'several', 'easy', 'view', 'cannot', 'extended', 'lauri', 'beaumont', 'union', 'times', 'open', 'cause', 'monthly', 'action', 'offers', 'industry', 'states', 'side', 'mailto', 'probably', 'neal', 'second', 'stephanie', 'download', 'flash', 'agree', 'mcf', 'transfer', 'doing', 'important', 'basis', 'different', 'final', 'koch', 'exxon', 'remove', 'microsoft', 'interested', 'application', 'sept', 'mg', 'write', 'lp', 'east', 'requirements', 'code', 'value', 'thank', 'together', 'exploration', 'mid', 'dfarmer', 'everything', 'receipt', 'thu', 'afternoon', 'late', 'enserch', 'coming', 'bank', 'response', 'tell', 'shipping', 'night', 'events', 'cynthia', 'lsp', 'close', 'legal', 'country', 'direct', 'expected', 'ces', 'corporation', 'options', 'really', 'voip', 'nominated', 'etc', 'latest', 'potential', 'priced', 'edward', 'valero', 'material', 'stack', 'victor', 'redeliveries', 'loss', 'remember', 'baumbach', 'option', 'private', 'longer', 'aware', 'included', 'drugs', 'public', 'reinhardt', 'version', 'hesse', 'discuss', 'related', 'asked', 'say', 'viagra', 'revision',

'bgcolor', 'kind', 'pro', 'completed', 'health', 'ready', 'plans', 'registered', 'regards', 'carthage', 'zone', 'fill', 'away', 'computer', 'systems', 'industrial', 'mentioned', 'told', 'therefore', 'growth', 'sold', 'track', 'reports', 'south', 'rd', 'jim', 'costs', 'image', 'expect', 'return', 'physical', 'el', 'browser', 'donna', 'stacey', 'begin', 'china', 'duty', 'approximately', 'showing', 'unit', 'jones', 'hard', 'verify', 'updated', 'eol', 'cs', 'orders', 'talk', 'trying', 'base', 'given', 'server', 'source', 'pathed', 'strong', 'bryan', 'directly', 'risks', 'whole', 'major', 'users', 'purchases', 'oo', 'karen', 'luong', 'level', 'required', 'delivered', 'portfolio', 'riley', 'ali', 'easttexas', 'poorman', 'bellamy', 'assistance', 'nothing', 'gif', 'thing', 'retail', 'didn', 'valley', 'department', 'cleburne', 'allow', 'gpgfin', 'answer', 'items', 'paste', 'avila', 'taken', 'mm', 'nguyen', 'ensure', 'reference', 'hall', 'later', 'lone', 'user', 'methanol', 'facility', 'network', 'spoke', 'though', 'tabs', 'taking', 'status', 'considered', 'purchased', 'says', 'yourself', 'paliourg', 'dy', 'jeff', 'businesses', 'fred', 'transportation', 'apache', 'morris', 'nov', 'ltd', 'brand', 'federal', 'statement', 'oasis', 'reflect', 'assets', 'lamadrid', 'general', 'bridge', 'ability', 'oct', 'play', 'enrononline', 'compliance', 'spam', 'availability', 'king', 'understanding', 'chance', 'quick', 'effort', 'points', 'reliantenergy', 'fixed', 'short', 'hill', 'cheryl', 'aepin', 'key', 'understand', 'valign', 'capacity', 'game', 'took', 'bring', 'guys', 'god', 'green', 'care', 'withers', 'property', 'hub', 'johnson', 'employee', 'wants', 'albrecht', 'meaning', 'expectations', 'mx', 'moved', 'cernosek', 'matter', 'devon', 'calls', 'worldwide', 'records', 'removed', 'lose', 'large', 'referenced', 'walker', 'iferc', 'enw', 'ponton', 'eileen', 'ship', 'upon', 'enerfin', 'jennifer', 'looks', 'staff', 'pc', 'target', 'waha', 'making', 'cp', 'impact', 'partner', 'immediately', 'shall', 'channel', 'takes', 'sat', 'others', 'hear', 'went', 'travel', 'listing', 'approved', 'processing', 'early', 'enough', 'sally', 'starting', 'distribution', 'tejas', 'transactions', 'stay', 'earl', 'superty', 'doesn', 'reserves', 'includes', 'choose', 'adobe', 'publisher', 'paso', 'cornhusker', 'training', 'markets', 'content', 'solution', 'shell', 'jpg', 'print', 'drive', 'pain', 'password', 'half', 'herrera', 'saturday', 'moopid', 'hotlist', 'balance', 'super', 'vacation', 'sex', 'happy', 'excess', 'existing', 'fund', 'stella', 'share', 'sign', 'wells', 'won', 'four', 'text', 'card', 'tisdale', 'fwd', 'appreciate', 'non', 'experience', 'savings', 'settlements', 'draft', 'couple', 'informed', 'biz', 'watch', 'plus', 'sun', 'expense', 'images', 'land', 'occur', 'flowing', 'mar', 'terry', 'darren', 'cheap', 'weight', 'dynegy', 'activities', 'become', 'mr', 'format', 'attention', 'entire', 'photoshop', 'williams', 'instructions', 'neon', 'janet', 'contains', 'ago', 'friends', 'against', 'boas', 'music', 'certain', 'liz', 'svcs', 'record', 'fast', 'dave', 'held', 'mind', 'ua', 'publication', 'differ', 'comments', 'fun', 'rest', 'instant', 'agent', 'communications', 'director', 'partners', 'investors', 'expedia', 'kevin', 'assist', 'safe', 'approval', 'allocate', 'black', 'none', 'intrastate', 'document', 'eric', 'hakemack', 'expired', 'lower', 'active', 'secure', 'cc', 'five', 'determine', 'press', 'colspan', 'missing', 'jill', 'discussion', 'relief', 'respect', 'specific', 'technologies', 'al', 'holmes', 'white', 'yesterday', 'medical', 'pinion', 'sorry', 'men', 'leave', 'pass', 'video', 'gomes', 'doctor', 'projects', 'limit', 'air', 'knle', 'pharmacy', 'confirmation', 'opportunity', 'involve', 'notify', 'gtc', 'class', 'ken', 'started', 'outage', 'confidential', 'room', 'blue', 'estimated', 'officer', 'reach', 'messages', 'database', 'words', 'prc', 'tracked', 'transition', 'light', 'national', 'hot', 'offering', 'gulf', 'provides', 'iit', 'demokritos', 'mckay', 'average', 'wide', 'heard', 'files', 'dan', 'billed', 'mccoy', 'rc', 'exactly', 'middle', 'select', 'bruce', 'louisiana', 'receiving', 'california', 'event', 'roll', 'mops', 'william', 'appear', 'perfect', 'html', 'features', 'join', 'greater', 'sunday', '

pick', 'featured', 'cdnow', 'prize', 'reveffo', 'olsen', 'expects', 'estimate', 'near', 'common', 'package', 'title', 'whether', 'bought', 'evergreen', 'difference', 'elizabeth', 'history', 'monitor', 'advised', 'result', 'sources', 'school', 'unaccounted', 'paragraph', 'turn', 'kimberly', 'increased', 'communication', 'members', 'concerns', 'uncertainties', 'associated', 'reduce', 'committed', 'wi', 'asap', 'goes', 'trader', 'waiting', 'canada', 'worth', 'representative', 'claim', 'ceo', 'london', 'discussions', 'php', 'brazos', 'trevino', 'calling', 'involved', 'la', 'gift', 'southern', 'groups', 'hour', 'tufco', 'previously', 'voice', 'normally', 'resolve', 'efforts', 'nor', 'recent', 'purchasing', 'county', 'ok', 'express', 'generic', 'according', 'respond', 'situation', 'hold', 'lot', 'interconnect', 'word', 'came', 'west', 'role', 'opportunities', 'corporate', 'remain', 'similar', 'readers', 'suggestions', 'subscribers', 'projections', 'lead', 'learn', 'resolved', 'agreed', 'sec', 'head', 'enjoy', 'img', 'rnd', 'responsible', 'outstanding', 'member', 'panenergy', 'american', 'cass', 'register', 'promotions', 'parties', 'winfree', 'selling', 'usage', 'appropriate', 'assignment', 'media', 'believes', 'require', 'submit', 'model', 'spinnaker', 'copano', 'facilities', 'opinion', 'factors', 'identified', 'beverly', 'ews', 'gdp', 'deliver', 'job', 'profile', 'across', 'neuweiler', 'suggest', 'girls', 'manage', 'usa', 'local', 'bad', 'greg', 'vs', 'fees', 'digital', 'cf', 'strangers', 'registration', 'delta', 'rolex', 'goliad', 'hesco', 'success', 'primary', 'quarter', 'course', 'chairman', 'petroleum', 'notes', 'medications', 'ei', 'instead', 'fine', 'lake', 'pre', 'force', 'seek', 'recipient', 'gain', 'placed', 'age', 'least', 'body', 'asking', 'discussed', 'hanson', 'emails', 'nominate', 'ext', 'known', 'ones', 'ed', 'assigned', 'htmlimg', 'means', 'present', 'various', 'invoices', 'gd', 'agency', 'along', 'located', 'reflects', 'solutions', 'ex', 'house', 'cds', 'br', 'owner', 'apr', 'sullivan', 'basin', 'linda', 'worked', 'car', 'seen', 'properties', 'booked', 'higher', 'store', 'est', 'revenue', 'wait', 'women', 'far', 'met', 'wholesale', 'range', 'kcs', 'recorded', 'brown', 'lots', 'match', 'input', 'grant', 'providing', 'huge', 'investor', 'kelly', 'apply', 'paths', 'handling', 'pipes', 'advantage', 'analysis', 'focus', 'draw', 'red', 'origination', 'connection', 'planning', 'wilson', 'golf', 'summary', 'item', 'bankruptcy', 'expenses', 'pgev', 'encina', 'beaty', 'memo', 'initial', 'thousand', 'mills', 'penis', 'friend', 'conversation', 'multiple', 'martin', 'names', 'bit', 'dth', 'talked', 'behalf', 'preliminary', 'button', 'herein', 'gisb', 'coupon', 'sa', 'oi', 'appears', 'door', 'texaco', 'csikos', 'arrangements', 'cpr', 'expires', 'popular', 'sending', 'research', 'conditions', 'gb', 'board', 'ca', 'applications', 'tried', 'paying', 'acquisition', 'reporting', 'normal', 'maintenance', 'resume', 'announced', 'attachment', 'buyer', 'objectives', 'prod', 'represent', 'sandi', 'hplnl', 'government', 'committee', 'running', 'tetco', 'discount', 'jo', 'holding', 'earlier', 'positions', 'happen', 'mailing', 'decided', 'recently', 'chris', 'xanax', 'valium', 'broadband', 'individual', 'station', 'td', 'financing', 'somehow', 'pena', 'critical', 'attend', 'kristen', 'inform', 'highly', 'hl', 'phillips', 'minutes', 'titles', 'affiliate', 'wife', 'lonestar', 'charlotte', 'quickly', 'paper', 'test', 'comes', 'mobile', 'internal', 'privacy', 'ideas', 'live', 'gotten', 'floor', 'benefit', 'percent', 'ms', 'dr', 'ebs', 'msn', 'gave', 'dallas', 'enterprise', 'rx', 'spring', 'ftar', 'ooking', 'hawkins', 'exclusive', 'selected', 'baxter', 'actually', 'single', 'shop', 'nominates', 'guarantee', 'minute', 'correctly', 'unique', 'bid', 'building', 'stated', 'accept', 'assumptions', 'centana', 'senior', 'pill', 'kinsey', 'sap', 'immediate', 'goals', 'category', 'mitchell', 'acceptance', 'termination', 'sweeney', 'facts', 'amazon', 'arrangement', 'josey', 'funds', 'among', 'accuracy', 'mean', 'rather', 'kim', 'egmnom', 'indicate', 'updates', 'extra', 'adjustment', 'accounts', 'lowest', 'gold', 'purposes', 'remaining', 'talking', 'entry', 'road', 'load', 'simply', 'europe', 'lindley', 'unders

tood', 'logos', 'hi', 'speed', 'profit', 'notified', 'jackson', 'z', 'vols', 'serve', 'additionally', 'shipped', 'connor', 'fontfont', 'q', 'kept', 'dollar', 'jr', 'almost', 'fri', 'paul', 'documents', 'analyst', 'crude', 'cap', 'shopping', 'aug', 'clearance', 'schneider', 'ftworth', 'father', 'anticipated', 'resellers', 'congress', 'counterparties', 'epgt', 'buying', 'san', 'invest', 'cartwheel', 'brandywine', 'wrong', 'mtbe', 'split', 'submitted', 'hull', 'gra', 'children', 'leader', 'true', 'baseload', 'mb', 'letters', 'billion', 'rights', 'mtr', 'heidi', 'clean', 'historical', 'asset', 'foreign', 'gr', 'entity', 'developed', 'maybe', 'jeffrey', 'transmission', 'outside', 'lost', 'membership', 'invitation', 'ocean', 'legislation', 'hernandez', 'pep', 'payments', 'wallis', 'rev', 'kenneth', 'seaman', 'annual', 'guess', 'bammel', 'lines', 'guadalupe', 'zivley', 'exception', 'example', 'pathing', 'revisions', 'pipelines', 'equity', 'budget', 'wed', 'dealers', 'window', 'juno', 'claims', 'bottom', 'standard', 'alternative', 'merchant', 'braband', 'topica', 'telephone', 'reliant', 'speculative', 'yes', 'en', 'morgan', 'cable', 'edmondson', 'participate', 'usb', 'throughout', 'checked', 'myself', 'contents', 'fat', 'investments', 'six', 'build', 'giving', 'calendar', 'inherent', 'edition', 'darial', 'hr', 'trip', 'pull', 'moving', 'concern', 'propose', 'rm', 'deer', 'enquiries', 'alt', 'tammy', 'front', 'reduction', 'evening', 'concerning', 'gets', 'effect', 'isn', 'haven', 'cowboy', 'sea', 'dvd', 'launch', 'minimum', 'changing', 'built', 'avoid', 'chief', 'stephen', 'chad', 'manual', 'finally', 'strategy', 'executive', 'thousands', 'conflict', 'resulting', 'policy', 'commission', 'stand', 'positive', 'quantity', 'programs', 'airmail', 'texoma', 'prepared', 'austin', 'matt', 'intent', 'uae', 'citibank', 'jaquet', 'hol', 'harris', 'min', 'hplr', 'advance', 'weather', 'terminated', 'whom', 'sheet', 'venturatos', 'cellpadding', 'hotel', 'leading', 'guaranteed', 'idea', 'announce', 'pleased', 'award', 'operational', 'prepare', 'schedulers', 'child', 'sum', 'quote', 'adjusted', 'warning', 'issued', 'ga', 'cross', 'detail', 'pertaining', 'tess', 'owe', 'crow', 'availabilities', 'griffin', 'christy', 'crosstex', 'eel', 'itoy', 'heart', 'licensed', 'overnight', 'cal', 'otherwise', 'luck', 'stretch', 'generation', 'broker', 'construed', 'except', 'traders', 'carry', 'column', 'approx', 'main', 'alert', 'charges', 'step', 'revenues', 'games', 'gottlob', 'looked', 'individuals', 'beck', 'stuff', 'welcome', 'port', 'glover', 'description', 'daniel', 'quantities', 'park', 'managing', 'town', 'seller', 'summer', 'tina', 'dates', 'eff', 'dudley', 'ferc', 'robin', 'charles', 'customerservice', 'zonedubai', 'emirates', 'aeor', 'clickathome', 'materia', 'island', 'vaughn', 'sexual', 'eiben', 'forms', 'delete', 'realize', 'tailgate', 'behind', 'villarreal', 'lon', 'benoit', 'simple', 'tech', 'ahead', 'double', 'ordering', 'se', 'miss', 'law', 'eb', 'post', 'outlook', 'equipment', 'leslie', 'reeves', 'org', 'tools', 'cold', 'adjustments', 'contained', 'saw', 'edit', 'deciding', 'finance', 'patti', 'listbot', 'river', 'kathryn', 'holiday', 'successful', 'unable', 'advisor', 'pool', 'bryce', 'outages', 'adjust', 'screen', 'otc', 'brent', 'helps', 'auto', 'foot', 'region', 'links', 'contain', 'knowledge', 'yvette', 'dial', 'pressure', 'detailed', 'indicated', 'charged', 'sites', 'make', 'female', 'mcmills', 'cook', 'mazowita', 'meredith', 'allocations', 'meetings', 'particular', 'environment', 'drug', 'search', 'mailings', 'designed', 'rock', 'measurement', 'art', 'corrected', 'kids', 'benefits', 'tv', 'seems', 'husband', 'fix', 'grow', 'decision', 'wireless', 'mo', 'conference', 'interview', 'levels', 'copies', 'cindy', 'urgent', 'regular', 'payroll', 'shown', 'consumers', 'reliable', 'tr', 'indicating', 'coast', 'greif', 'severson', 'tri', 'vicodin', 'liquids', 'significant', 'intend', 'usd', 'pager', 'avails', 'spencer', 'ce', 'charset', 'verdana', 'fully', 'flynn', 'da', 'personnel', 'multi', 'closed', 'vice', 'administration', 'gmt', 'midstream', 'eye', 'speckels', 'studio', 'cilco', 'likely', 'managers', 'structure', 'sit', 'parent', 'preparation', 'mix', 'mmbtus', 'timing', 'happening', 'lottery'

, 'killing', 'acquire', 'mack', 'pcx', 'fares', 'international', 'notification', 'swift', 'identify', 'areas', 'separate', 'unless', 'producers', 'allows', 'pretty', 'waste', 'joanie', 'drop', 'taxes', 'premium', 'teams', 'choice', 'largest', 'addressed', 'dolphin', 'ngo', 'self', 'davis', 'htm', 'ad', 'graphics', 'hit', 'competitive', 'thus', 'incorrect', 'ti', 'acts', 'previous', 'edu', 'proven', 'electric', 'pictures', 'charlene', 'benedict', 'chevron', 'treatment', 'lesson', 'player', 'sds', 'wc', 'intraday', 'assurance', 'sdsnom', 'rebecca', 'quit', 'netco', 'intra', 'whatever', 'lyondell', 'reviewed', 'solicitation', 'filings', 'log', 'noon', 'locations', 'joe', 'completely', 'rivers', 'language', 'street', 'automatically', 'ft', 'powerful', 'specials', 'alone', 'fyi', 'properly', 'proper', 'explode', 'decrease', 'medication', 'desks', 'impacted', 'anywhere', 'completion', 'banking', 'consider', 'certificate', 'exercise', 'zeroed', 'websites', 'tonight', 'diligence', 'education', 'club', 'vegas', 'affordable', 'sports', 'predictions', 'billing', 'diamond', 'posted', 'prayer', 'actions', 'nomad', 'results', 'jason', 'purpose', 'deposit', 'entertainment', 'materially', 'blank', 'resolution', 'anderson', 'nat', 'rom', 'soma', 'organization', 'aquila', 'solid', 'affected', 'transco', 'spend', 'responsibilities', 'assume', 'header', 'accountant', 'functionality', 'meant', 'killed', 'analysts', 'rick', 'rolled', 'noted', 'discovered', 'offices', 'torch', 'often', 'york', 'joint', 'briley', 'competition', 'guide', 'intercompany', 'son', 'settlement', 'presently', 'cart', 'tim', 'entries', 'russ', 'valadez', 'rules', 'molly', 'apple', 'atleast', 'scheduler', 'pi', 'hector', 'dell', 'opm', 'hottlist', 'yap', 'gone', 'heal', 'llc', 'setting', 'reached', 'proposal', 'hundred', 'trust', 'official', 'table', 'mcgee', 'written', 'operation', 'cellspacing', 'laptop', 'feature', 'ram', 'victoria', 'larry', 'units', 'requests', 'continued', 'external', 'pack', 'couldn't', 'lateral', 'strictly', 'resource', 'although', 'sr', 'commodity', 'pulled', 'protocol', 'bed', 'generated', 'redmond', 'girl', 'apparently', 'tool', 'reviews', 'released', 'movies', 'inside', 'shareholder', 'rr', 'compensation', 'beliefs', 'foresee', 'lease', 'rule', 'marta', 'chemical', 'hillary', 'hp', 'tongue', 'adonis', 'advises', 'master', 'eight', 'wasn't', 'itself', 'documentation', 'xl', 'humble', 'elsa', 'pics', 'hughes', 'brokered', 'distribute', 'consultation', 'sheri', 'lists', 'cannon', 'treated', 'factor', 'putting', 'verified', 'releases', 'enhanced', 'controls', 'craig', 'worksheet', 'conversion', 'max', 'hrs', 'helpful', 'hand', 'producing', 'dl', 'developing', 'design', 'woman', 'understands', 'standards', 'promotion', 'sarco', 'hospital', 'ffffff', 'respective', 'richmond', 'conoco', 'driver', 'easily', 'sean', 'den', 'gateway', 'holdings', 'brad', 'college', 'gains', 'adult', 'dated', 'em', 'mcloughlin', 'anticipates', 'henderson', 'julia', 'negotiations', 'sofftwaares', 'garrick', 'comstock', 'trochta', 'imceanotes', 'ecom', 'larger', 'nommensen', 'coordinate', 'partnership', 'otcbb', 'announces', 'louis', 'dealer', 'reliance', 'season', 'agua', 'dulce', 'offshore', 'gathered', 'forever', 'function', 'happened', 'sample', 'easier', 'aim', 'pa', 'expensive', 'thinks', 'maximum', 'war', 'mining', 'drilling', 'owned', 'todd', 'advanced', 'provider', 'pending', 'providers', 'silver', 'cherry', 'hundreds', 'thoughts', 'addresses', 'beach', 'baby', 'requires', 'caused', 'variance', 'extension', 'carbide', 'anytime', 'adding', 'triple', 'dawn', 'martinez', 'entering', 'login', 'bretz', 'ls', 'writeoff', 'locker', 'wiil', 'block', 'blood', 'romeo', 'responsibility', 'brennan', 'btu', 'venture', 'connected', 'nascar', 'opinions', 'executed', 'cell', 'flag', 'doctors', 'invoiced', 'marlin', 'coffey', 'nice', 'amazing', 'ii', 'determined', 'handled', 'keeping', 'touch', 'upgrade', 'shipment', 'brought', 'forwarding', 'confidence', 'hesitate', 'seem', 'electronic', 'appreciated', 'deadline', 'franklin', 'heather', 'reasons', 'passed', 'safety', 'procedures', 'payback', 'networks', 'utility', 'count', 'africa', 'exact', 'creating', 'loading', 'processed', 'court', 'tier', 'sender', 'att', 'mailbox', 'glad', 'buddy', 'prof

iles', 'portion', 'protection', 'compressor', 'okay', 'oba', 'finding', 'heads', 'bar', 'turned', 'remote', 'illustrator', 'oem', 'noticed', 'mails', 'darron', 'nick', 'urbanek', 'jerry', 'barrett', 'ehronline', 'und', 'abdv', 'egm', 'coud', 'technoiogies', 'owns', 'improved', 'eat', 'moment', 'owners', 'develop', 'installed', 'videos', 'frank', 'hearing', 'inches', 'busy', 'ref', 'valuable', 'et', 'un', 'url', 'shawna', 'iso', 'capture', 'extremely', 'ya', 'causing', 'consent', 'anyway', 'round', 'discrepancies', 'cheapest', 'confidentiality', 'disclosure', 'prohibited', 'vol', 'correction', 'communicate', 'processes', 'spain', 'shareholders', 'supported', 'smoking', 'mine', 'biggest', 'erections', 'platform', 'miles', 'exciting', 'association', 'die', 'restricted', 'ma', 'income', 'goal', 'bane', 'collection', 'nathan', 'wind', 'piece', 'familiar', 'gore', 'experiencing', 'pico', 'mai', 'dewpoint', 'tessie', 'hair', 'bussell', 'diane', 'delivering', 'originally', 'accurate', 'began', 'seven', 'tracking', 'randall', 'gay', 'emerging', 'prescriptions', 'story', 'arial', 'florida', 'space', 'ownership', 'european', 'sutton', 'concerned', 'male', 'spent', 'agreements', 'industries', 'picture', 'filled', 'continues', 'death', 'choate', 'majeure', 'device', 'hence', 'ten', 'campaign', 'massive', 'eyes', 'requesting', 'lives', 'reminder', 'eliminate', 'copied', 'consemtu', 'died', 'sound', 'offered', 'expressed', 'anti', 'duplicate', 'steps', 'books', 'improve', 'implementation', 'gives', 'ac', 'peggy', 'proprietary', 'ways', 'advertisement', 'published', 'earnings', 'mortgage', 'consumer', 'ct', 'tape', 'fl', 'cia', 'organizational', 'agenda', 'rental', 'carriere', 'moshou', 'church', 'trouble', 'medium', 'aggressive', 'smart', 'zajac', 'ail', 'participants', 'gap', 'earthlink', 'wire', 'trades', 'messaging', 'ut', 'wil', 'richardson', 'blvd', 'glo', 'seneca', 'publiisher', 'imited', 'isc', 'contacts', 'sleep', 'kyle', 'cooperation', 'possibly', 'leaving', 'motor', 'hopefully', 'tie', 'speak', 'mi', 'suggested', 'canadian', 'uses', 'connect', 'pvr', 'rich', 'places', 'auction', 'po', 'spacer', 'client', 'recommended', 'royalty', 'amended', 'default', 'living', 'regardless', 'human', 'bringing', 'focused', 'stores', 'variety', 'netherlands', 'leaders', 'bowen', 'salary', 'signed', 'penny', 'loan', 'desktop', 'chase', 'pleasure', 'compare', 'session', 'overall', 'stranger', 'length', 'planned', 'sp', 'darrel', 'raise', 'palestinian', 'expiration', 'serial', 'premiere', 'suzanne', 'reduced', 'players', 'applicable', 'impotence', 'buckley', 'wayne', 'hansen', 'indicative', 'sabrae', 'dating', 'winners', 'marshall', 'highest', 'ea', 'presentation', 'allowed', 'square', 'danny', 'gepl', 'hydrocarbon', 'alpine', 'christmas', 'muscle', 'souza', 'relating', 'begins', 'ecf', 'forth', 'answers', 'audit', 'approve', 'lunch', 'types', 'starts', 'difficult', 'le', 'lasts', 'series', 'till', 'edge', 'growing', 'covered', 'shipper', 'sometime', 'republic', 'filter', 'sooner', 'increasing', 'nelson', 'percentage', 'returned', 'pop', 'interface', 'kin', 'experienced', 'prime', 'merger', 'obtain', 'ryan', 'servers', 'attachments', 'achieve', 'effects', 'gov', 'examples', 'procedure', 'explore', 'caribbean', 'rally', 'amounts', 'comfort', 'attempt', 'greatly', 'amelia', 'engel', 'delay', 'fare', 'der', 'cove', 'filing', 'fletcher', 'leth', 'undervalued', 'cents', 'esther', 'hlavaty', 'reid', 'lls', 'troy', 'palmer', 'metals', 'las', 'carter', 'luis', 'migration', 'brief', 'hess', 'therein', 'ur', 'pond', 'joanne', 'community', 'tglo', 'eogi', 'ml', 'wysak', 'felipe', 'errors', 'affect', 'convenient', 'minimal', 'boost', 'incremental', 'decide', 'reserve', 'superior', 'kerr', 'willing', 'quite', 'wild', 'unlimited', 'sans', 'mother', 'computers', 'unfortunately', 'ordered', 'satisfaction', 'priority', 'traded', 'testing', 'portal', 'ward', 'lets', 'aren', 'knows', 'refer', 'shot', 'fda', 'tue', 'saying', 'cancel', 'forecast', 'cousino', 'bass', 'permanent', 'phones', 'technical', 'whose', 'objective', 'cards', 'distributed', 'learning', 'fire', 'drill', 'towards', 'forget', 'explosion', 'gloria', 'formula', 'redelivery', 'audio', 'visual', 'encoding', 'approach', 'doubt', 'staffing', 'excite', 'corel', 'tm', 'e

```
nronavailso', 'contacting', 'alland', 'heavy', 'economic', 'nigeria', 'milwa
ukee', 'phillip', 'curve', 'returns', 'padre', 'kathy', 'buttons', 'sir', 'v
ary', 'sounds', 'disclose', 'authority', 'flw', 'straight', 'worldnet', 'bee
mer', 'ooo', 'defs', 'thorough', 'officers', 'flight', 'prefer', 'awesome',
'macintosh', 'feet', 'constitutes', 'formosa', 'porn', 'armstrong', 'driscol
l', 'watches', 'newsietter', 'twenty', 'tommy', 'fields', 'method', 'setup',
'allocating', 'initially', 'missed', 'clarification', 'especially', 'dorcheu
s', 'del', 'millions', 'insurance', 'pooling', 'trial', 'tennessee', 'ellis'
, 'direction', 'bold', 'catch', 'performing', 'accepted', 'matters', 'batch'
, 'continuing', 'winning', 'symbol', 'offsystem', 'decisions', 'produced', '
ended', 'greatest', 'degree', 'solmonson', 'imbalances', 'fall', 'fear', 'ha
te', 'fight', 'reallocated', 'debt', 'reform', 'australia', 'plain', 'prompt
', 'remains', 'ifhsc', 'enhancements', 'connevey', 'jay', 'valued', 'lay', '
infrastructure', 'military', 'allowing', 'ff', 'dry', 'Prediction']
```

Data Types:

```
Email No.      str
the            int64
to            int64
ect           int64
and           int64
...
military      int64
allowing      int64
ff           int64
dry          int64
Prediction    int64
Length: 3002, dtype: object
```

Memory information

```
<class 'pandas.DataFrame'>
RangeIndex: 5172 entries, 0 to 5171
Columns: 3002 entries, Email No. to Prediction
dtypes: int64(3001), str(1)
memory usage: 118.5 MB
First 5 elements
```

	Email No.	the	to	ect	and	for	of	a	you	hou	...	connevey	jay
0	Email 1	0	0	1	0	0	0	2	0	0	...	0	0
1	Email 2	8	13	24	6	6	2	102	1	27	...	0	0
2	Email 3	0	0	1	0	0	0	8	0	0	...	0	0
3	Email 4	0	5	22	0	5	1	51	2	10	...	0	0
4	Email 5	7	6	17	1	5	2	57	0	9	...	0	0

	valued	lay	infrastructure	military	allowing	ff	dry	Prediction
0	0	0		0	0	0	0	0
1	0	0		0	0	0	1	0
2	0	0		0	0	0	0	0
3	0	0		0	0	0	0	0
4	0	0		0	0	0	1	0

[5 rows x 3002 columns]

	missing	percent
Email No.	0	0.0
the	0	0.0
to	0	0.0

```

ect          0      0.0
and          0      0.0
...          ...    ...
military     0      0.0
allowing     0      0.0
ff           0      0.0
dry          0      0.0
Prediction   0      0.0

```

[3002 rows x 2 columns]
There is no missing value!

The duplicate rows are : 0
No duplicates detected!

Summary statistics

	the	to	ect	and	for \
count	5172.000000	5172.000000	5172.000000	5172.000000	5172.000000
mean	6.640565	6.188128	5.143852	3.075599	3.124710
std	11.745009	9.534576	14.101142	6.045970	4.680522
min	0.000000	0.000000	1.000000	0.000000	0.000000
25%	0.000000	1.000000	1.000000	0.000000	1.000000
50%	3.000000	3.000000	1.000000	1.000000	2.000000
75%	8.000000	7.000000	4.000000	3.000000	4.000000
max	210.000000	132.000000	344.000000	89.000000	47.000000

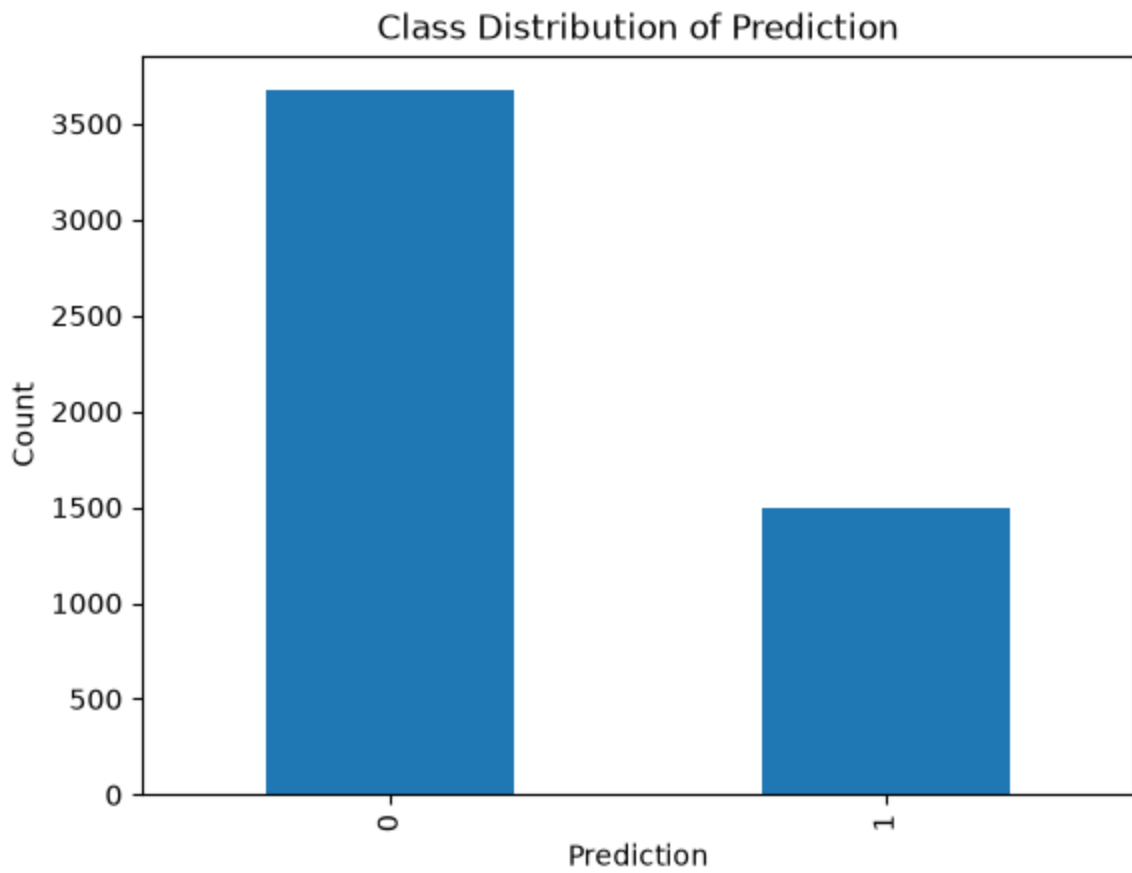
	of	a	you	hou	in ...
count	5172.000000	5172.000000	5172.000000	5172.000000	5172.000000 ...
mean	2.627030	55.517401	2.466551	2.024362	10.600155 ...
std	6.229845	87.574172	4.314444	6.967878	19.281892 ...
min	0.000000	0.000000	0.000000	0.000000	0.000000 ...
25%	0.000000	12.000000	0.000000	0.000000	1.000000 ...
50%	1.000000	28.000000	1.000000	0.000000	5.000000 ...
75%	2.000000	62.250000	3.000000	1.000000	12.000000 ...
max	77.000000	1898.000000	70.000000	167.000000	223.000000 ...

	connevey	jay	valued	lay	infrastructure \
count	5172.000000	5172.000000	5172.000000	5172.000000	5172.000000
mean	0.005027	0.012568	0.010634	0.098028	0.004254
std	0.105788	0.199682	0.116693	0.569532	0.096252
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000	0.000000
max	4.000000	7.000000	2.000000	12.000000	3.000000

	military	allowing	ff	dry	Prediction
count	5172.000000	5172.000000	5172.000000	5172.000000	5172.000000
mean	0.006574	0.004060	0.914733	0.006961	0.290023
std	0.138908	0.072145	2.780203	0.098086	0.453817
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	1.000000	0.000000	1.000000
max	4.000000	3.000000	114.000000	4.000000	1.000000

[8 rows x 3001 columns]

Class Distribution



```
Out[27]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>Email No.</th> <th>the</th> <th>to</th> <th>ect</th> <th>and</th>
<th>for</th> <th>of</th> <th>a</th> <th>you</th> <th>hou</th> <th>...</th>
<th>connevey</th> <th>jay</th> <th>valued</th> <th>lay</th> <th>infrastructure</th>
<th>military</th> <th>allowing</th> <th>ff</th> <th>dry</th> <th>Prediction</th> </
thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th>
<th>5167</th> <th>5168</th> <th>5169</th> <th>5170</th> <th>5171</th> </tbody>
<tbody></tbody>
```

Email 1	0	0	1	0	0	0	2	0	0	...	0	0	0	0	0	0	0	0	0	0
Email 2	8	13	24	6	6	2	102	1	27	...	0	0	0	0	0	0	0	1	0	0
Email 3	0	0	1	0	0	0	8	0	0	...	0	0	0	0	0	0	0	0	0	0
Email 4	0	5	22	0	5	1	51	2	10	...	0	0	0	0	0	0	0	0	0	0
Email 5	7	6	17	1	5	2	57	0	9	...	0	0	0	0	0	0	0	1	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Email 5168	2	2	2	3	0	0	32	0	0	...	0	0	0	0	0	0	0	0	0	0
Email 5169	35	27	11	2	6	5	151	4	3	...	0	0	0	0	0	0	0	1	0	0
Email 5170	0	0	1	1	0	0	11	0	0	...	0	0	0	0	0	0	0	0	0	1
Email 5171	2	7	1	0	2	1	28	2	0	...	0	0	0	0	0	0	0	1	0	1
Email 5172	22	24	5	1	6	5	148	8	2	...	0	0	0	0	0	0	0	0	0	0

<p>5172 rows × 3002 columns</p>

In []: